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Invariant-Based Molecular Similarity Search: An Interactive Map and Empirical Validation of Geometric Descriptors for PubChem Conformers

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Abstract

Molecular similarity search underpins virtual screening in drug discovery but the dominant method, RMSD, requires explicit molecular alignment and is undefined for molecules of different sizes. This project develops and validates an alignment-free, molecular similarity search system based on geometric invariants Sorted Radial Distances (SRD) and Sorted Pairwise Distances (SPD). The system comprises an interactive browser-based molecular map visualising 1,223 conformers from PubChem in invariant space with eight configurable axis options, real-time molecule upload and nearest-neighbour search, together with a two-stage KDTree-based filtration pipeline achieving an average 82.7% search space reduction across atom count groups containing similar conformers. Lipschitz continuity of both invariants is validated empirically across 50 molecules and seven perturbation magnitudes to yield empirical Lipschitz constants of $K \approx 1.045$ for SRD and $K \approx 1.592$ for SPD. SPD distance is compared against RMSD across 28 molecular pairs for a moderate positive Pearson correlation ($r = 0.502$, n = 11 atom-count-matched pairs). The system is delivered as a single reproducible Python script and a standalone interactive HTML visualisation.

Statement of Ethical Compliance

Data Category: A

Participant Category: 0

I confirm that I have followed ethical guidelines for this project. All data used is publicly available and non-personal. No human participants were involved in the research. No ethical concerns beyond those addressed above have been identified.

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1. Introduction

Modern drug discovery depends on the ability to identify molecules whose three-dimensional shapes are similar to known active compounds. As public molecular databases have grown to unprecedented scales, the computational challenge of performing similarity search efficiently has become one of the defining problems of computational chemistry.

1.1 Motivation and Problem Statement

PubChem, the world's largest open repository of chemical information, currently contains approximately 943 million three-dimensional conformers across 122 million unique compounds (Chatterjee, 2025; Kim et al., 2023). With a naive exhaustive approach, O(N^2) pairwise comparisons on all 943 million conformers are computationally infeasible without an efficient pre-filtering strategy. The current standard to address this task is Root Mean Square Deviation (RMSD), which requires explicit molecular alignment before comparison, is undefined when comparing molecules of different atom counts and can cause discontinuous changes in the metric under perturbations.

This project addresses the limitations of RMSD by developing and validating an alignment-free, interactive molecular similarity search system based on geometric invariants which are mathematical descriptors that are unchanged under any combination of translation, rotation and reflection. The two invariants that are used are Sorted Radial Distances (SRD), which encodes the distances from each atom to the molecular centre in O(n) time, and Sorted

Pairwise Distances (SPD), which encodes all interatomic distances in $O(n^2)$ time. These two invariants form the basis of a two-stage coarse-to-fine filtration pipeline that reduces the conformer search space without requiring alignment and an interactive molecular map that plots conformers in invariant space. To address the instability of RMSD under perturbation, both invariants are validated for Lipschitz continuity which is a mathematical property guaranteeing that small structural changes produce proportionally small changes in the descriptor.

1.2 Research Questions

The project addresses and tackles the below four research questions. Each of these research questions are addressed explicitly in Section 5 (Testing and Evaluation).

RQ1 — Can sorted distance invariants (SRD/SPD) produce an interactive molecular map that preserves chemical intuitions about molecular similarity?

RQ2 — Are SRD and SPD Lipschitz-continuous under small atomic perturbations, and if so, what is the empirical Lipschitz constant?

RQ3 — How well does SPD distance correlate with RMSD for same-molecule conformer pairs versus different-molecule pairs?

RQ4 — How effectively does a two-stage SRD to SPD filtration pipeline reduce the conformer search space compared to exhaustive comparison?

1.3 Key Questions

The following definitions are relevant throughout the dissertation and provide a foundation for the general reader. Further detail on each including mathematical formalisms is provided in Section 2.

Molecule	A molecule is a stable arrangement of atoms held together by covalent bonds. In this project, molecules are represented as finite sets of atoms with three-dimensional Cartesian coordinates.
Conformer (physical definition)	A conformer is a stable three-dimensional configuration of a molecule that arises possibly due to some thermal vibration. Physically distinct conformers of the same molecule have small displacements in some or all its atoms. This definition is relevant to the Lipschitz continuity validation in Section 5.2.2.
Conformer (PubChem definition)	In PubChem, a conformer is a computationally generated three-dimensional molecular structure stored in a database. These are distinct database records, not necessarily physically occurring conformations (Bolton et al., 2011).
Isometry	An isometry is a distance-preserving transformation under any composition of translations, rotations and reflections (Anosova and Kurlin, 2025). Two molecular configurations are isometric if one can be obtained from the other by an isometry, meaning they represent the same physical structure viewed from a different orientation (Anosova and Kurlin, 2021).
Invariant	An invariant is a property of an object that remains unchanged under a specified class of transformations. In this project, invariants are unchanged under isometric transformations.

Isometry invariant	An isometry invariant is a function I such that I(C) = I(C') for all isometric configurations C and C'. NOTE: this property makes the invariant suitable for alignment-free similarity search.
Lipschitz continuity	Specifically for a molecular invariant, this mathematical concept guarantees that small changes in atomic coordinates produce proportionally small changes in the descriptor.
RMSD	Root Mean Square Deviation (RMSD) is the standard alignment-based molecular similarity metric. It measures the minimum average atomic displacement between two optimally aligned structures.

Table 1.1 - Key definitions used throughout this dissertation

1.4 Prior Work

The most directly related prior work is Chatterjee (2025) who applied SRD and SPD invariants to PubChem's full 943 million conformer dataset on the University of Liverpool's BARKLA HPC cluster with search space reductions of 60–75%. He also produced a static scatter plot as a proof of concept for invariant-based molecular maps. This project extends that work by replacing the static plot with a fully interactive browser-based map with empirically validated Lipschitz continuity for the first time and external validation against RMSD. The theoretical foundation for the invariants is derived from Anosova and Kurlin (2021) on isometry invariants for atomic point clouds).

1.5 Aims, Objectives and Requirements

The following aims were introduced in the original project proposal:

Aim 1. Develop a visual mapping system for molecular databases using SRD and SPD descriptors to generate invariant structural embeddings.

Aim 2. Quantify the stability and robustness of the geometric invariants by experimentally validating their Lipschitz continuity under small atomic perturbations.

Aim 3. Investigate equivalence relations among molecules using both geometric invariants and the relational axioms of reflexivity, symmetry, and transitivity.

Aim 4. Evaluate the effectiveness of SRD/SPD-based descriptors in identifying near-duplicates and structurally similar compounds relative to standard metrics like RMSD.

Aim 5 (optional). Explore generative models using GFlowNet to sample new molecular structures from learned descriptor manifolds.

Two aims were modified during the project based on supervisor feedback and practical constraints. Aim 5 was descoped entirely because GFlowNet integration was optional in the original proposal and was not pursued as the core aims were ambitious. The use of dimensionality reduction techniques (UMAP, t-SNE) which was originally proposed for aim 1, was dropped on the supervisor's recommendation because such methods are stochastic, may produce different outputs on identical inputs and produce axes with no geometric meaning, which directly undermines the motivation for using mathematically grounded invariants (Kurlin, V., 2025, written feedback on project proposal). The GEOM dataset, originally planned as a secondary data source, could not be processed due to format obstacles as described in Section 2.2.2.

The below list is an overview of the essential and desirable requirements for the project. They reflect a more specific finalised scope of the project with adjustments made during development in response to supervisor feedback and practical constraints. Changes from the original proposal are documented above.

	Requirement	Addressed in
E1	Parse and process molecular conformer data from PubChem via REST API	*Sections 3.2, 4.1*
E2	Compute SRD and SPD invariant vectors for each conformer	*Sections 3.4, 4.2*
E3	Implement a two-stage KDTree-based filtration pipeline for search space reduction	*Sections 3.5, 4.3*
E4	Produce an interactive invariant-based molecular map with configurable axes	*Sections 3.6, 4.4*
E5	Validate Lipschitz continuity of SRD and SPD empirically	*Sections 3.7.1, 4.6, 5.2.2*
E6	Compare SPD distance against RMSD across a balanced set of molecular pairs	*Sections 3.7.2, 4.7, 5.2.3*
D1	Implement molecule upload feature in the browser using live PubChem API calls	*Section 4.5*
D2	Implement nearest-neighbour search in the browser-based map	*Section 4.5*
D3	Process the GEOM dataset and compare DFT and MMFF94 invariant maps	*Not achieved — Section 4.8*
D4	Integrate GFlowNet for generative molecular sampling	*Not achieved — descoped*

Table 1.2- *Essential (E) and desirable (D) requirements with cross-references. Requirements D3 and D4 were not achieved and detailed reasons are documented in Sections 4.8 and 1.5 respectively.*

1.6 Report Structure

Section 2 (Background and Related Work) provides the theoretical and technical context for the project and covers molecular databases, geometric invariant theory, RMSD, KDTree-based search and force fields. Section 3 (Design) presents the architectural and design decisions underlying the system with justification for each choice. Section 4 (Implementation) describes the realisation of the design including decisions made during programming, problems encountered and how they were resolved. Chapter 5 (Testing and Evaluation) presents unit test methodology and evaluates the system against each of the four research questions introduced in 1.2, an equivalence relation analysis and a formal threats to validity assessment. Chapter 6 (Project Ethics) addresses the ethical considerations of the project. Chapter 7 (Conclusion and Future Work) summarises the contributions of the project and identifies directions for future research. Chapter 8 (BCS Criteria and Self-Reflection) maps the project against the BCS assessment criteria and reflects on the development process.

2. Background and Related Work

This section presents the context necessary to situate the project's contributions within the existing literature and is organised around the five pillars of the project: molecular databases and data formats; geometric invariant theory; the standard RMSD similarity metric and its limitations; efficient spatial search structures and molecular force fields.

2.1 Molecular Similarity in Drug Discovery

The search for new therapeutic compounds is one of the most computationally demanding challenges in modern science because drug discovery relies heavily on the principle of molecular similarity since compounds with similar three-dimensional structures tend to exhibit similar biological activity. This relationship is known as the similarity-property principle (Lipinski et al., 2004). Filtering of large molecular databases to identify candidate compounds with specific properties depends on the ability to rapidly and accurately assess structural similarity between millions or billions of candidate molecules and known active compounds (Dobson, 2004).

Traditional approaches to virtual screening use two-dimensional fingerprint-based similarity measures, such as the Tanimoto coefficient over Morgan fingerprints, which encode molecular topology but discard three-dimensional geometric information. However, three-dimensional similarity measures, which account for the spatial arrangement of atoms, provide a more physically grounded basis for comparing molecular shapes and are increasingly used in structure-based drug design (Bolton et al., 2011). At the scale of PubChem's 943 million conformers (Chatterjee, 2025), a similarity measure that is both geometrically accurate and computationally efficient is required. Existing alignment-based methods such as RMSD struggle to provide this so this project addresses the challenge by developing and validating an invariant-based molecular similarity search system that is alignment-free, computationally efficient and mathematically stable under small atomic perturbations.

2.2 Molecular Databases and Data Formats

2.2.1 PubChem

PubChem is the world's largest open repository of chemical information, maintained by the National Institutes of Health (NIH) of the United States (Kim et al., 2023). The database organises molecular information across substance records (SID), which represent individual submissions from contributing organisations, compound records (CID), which represent unique chemical structures extracted and deduplicated from Substance records and conformer records, identified by a hexadecimal Conformer ID. As of the December 2025, PubChem contains approximately 943 million conformers across 122 million unique compounds (Chatterjee, 2025).

Three-dimensional conformers in PubChem are generated computationally using the MMFF94 force field (Tosco et al., 2014) and distributed as Structure Data Format (SDF) files. Access to the database is available via a REST API that supports targeted retrieval of specific compounds and conformers by name or identifier and a bulk FTP server providing compressed SDF archives for the entire dataset. PubChem's MMFF94 conformers are subject to generation constraints where only molecules with at most 50 non-hydrogen atoms, at most 15 rotatable bonds and supported element types are included (Bolton et al., 2011). While this restricts the chemical scope of the database, it ensures that all available conformers represent physically plausible three-dimensional geometries.

2.2.2 GEOM

The GEOM dataset (Axelrod and Gómez-Bombarelli, 2022) provides energy-annotated molecular conformations, generated using Density Functional Theory (DFT) which is a quantum mechanical method that offers substantially higher geometric accuracy than force field approaches such as MMFF94. GEOM contains approximately 37 million conformers for 450,000 unique molecules, with energies computed at the B97-D3 level of theory which makes them a more physically accurate benchmark for similarity search than PubChem's MMFF94 structures. However, GEOM was not successfully processed in this project due to format obstacles. The dataset distributes conformer data as serialised RDKit RdMol objects in Python pickle format, which proved incompatible with the SDF-based parsing pipeline developed for PubChem. This obstacle was anticipated in the project proposal and was identified by Chatterjee (2025) as a known limitation because it requires a custom parsing pipeline which was beyond the scope of this study.

2.2.3 Structured Data Format (SDF)

The Structure Data Format (SDF) is a text-based file format for representing chemical structural information (Bolton et al., 2011). Each SDF record contains a header block identifying the compound, a connection table (CTAB) comprising an atom block specifying the three-dimensional Cartesian coordinates and element symbol of each atom, a bond block specifying covalent connectivity and a properties block containing auxiliary information as key-value pairs.

2.3 Geometric Invariants

A central challenge in molecular similarity search is that the same molecule can be represented in infinitely many orientations so similarity measure that depends on the raw coordinates is therefore unsuitable for database search. The solution is to use geometric invariants which are functions of a molecular configuration that are unchanged under any isometric transformation.

2.3.1 Isometry

In Euclidean space $\mathbb{R}^3$, an isometry is a distance-preserving transformation under any composition of translations, rotations and reflections. Two molecular configurations C and C' are isometric if one can be obtained from the other by an isometry, meaning they represent the same physical structure (Anosova and Kurlin, 2025). An isometry invariant is a function I such that $I(C) = I(C')$ for all isometric configurations C and C'. For a molecular similarity measure to be alignment-free, identical structures in any orientation must produce identical descriptor values. A complete invariant is one where there are no false positives and two non-isometric molecules cannot share identical descriptor values *(Anosova and Kurlin, 2021)*.

2.3.2 Sorted Pairwise Distance and Sorted Radial Distance

The Sorted Pairwise Distance (SPD) invariant represents a molecule as the vector of all unique pairwise Euclidean distances between atoms, sorted in ascending order (Kurlin, 2024). SPD is computationally more demanding that Sorted Radial Distance (SRD) and requires O(n^2) time and space for the $\frac{n(n-1)}{2}$ pairwise distance computations. However, SRD is incomplete so for any two molecular configurations where all atoms lie on a sphere of the same radius centred at the geometric centre, will produce identical SRD vectors regardless of the angular arrangement of atoms. For example, ammonia (NH_3) and any molecule with identical atom count and radial distances, albeit different bond angles, would be indistinguishable by SRD alone. On the other hand, SPD is generically complete for almost all molecular configurations. Although it is theoretically possible for two non-isometric molecules to share identical SPD vectors, such cases are vanishingly rare for real three-dimensional molecular structures (Widdowson and Kurlin, 2023). The sort order was specified as ascending by the project supervisor because the shortest pairwise distances correspond to stronger bonded atoms. SRD and SPD are described in detail in Section 3.4.

2.3.2 Simplexwise Centred Distribution

The Simplexwise Centred Distribution (SCD) is a provably complete invariant for molecular configurations in R^3 [Widdowson and Kurlin, 2023; Kurlin, 2026]. The SCD encodes not only pairwise distances but also higher-order geometric relationships between atom subsets to achieve more overall completeness at the cost of significantly higher computational complexity. Although, SCD was considered for this project, it was not implemented on the account that SPD is sufficiently complete for the purposes of the study. Implementing SCD as a third-stage filter is impractical but for edge cases where SPD is genuinely ambiguous, could be a beneficial addition as future work.

2.4 RMSD

Root Mean Square Deviation (RMSD) is the standard metric for quantifying structural similarity between molecular conformers in computational chemistry and virtual screening pipelines (Bolton et al., 2011). For two molecular configurations C = {p_1, ..., p_n} and C' = {q_1, ..., q_n} with the same atom count n, RMSD is defined as:

$$RMSD(C, C') = \min\{R, t\} \sqrt{\left(\frac{1}{n}\right) \sum_{\mathrm{i}} \left||p_{\mathrm{i}} - (Rq_{\mathrm{i}} + t)|\right|^2}$$

where R is a rotation matrix and t is a translation vector. The minimisation over all rotations and translations is equivalent to solving an optimisation problem, which finds the optimal rigid-body alignment of C' onto C. RDKit's GetBestRMS function is used in this project to internally compute the RMSD comparison experiment (see Section 4.7).

RMSD dominates molecular similarity search because it is physically interpretable so an RMSD of 0.5 Å corresponds to an average atomic displacement of half an Ångström. It is reproducible in research because it is widely implemented across cheminformatics software. PubChem itself uses RMSD as a criterion for conformer diversity during conformer generation to cluster candidate conformers by pairwise RMSD to retain only the most diverse representations (Bolton et al., 2011).

However, RMSD has significant limitations that motivate the invariant-based approach of this project. First and foremost, RMSD requires explicit molecular alignment before the metric can be evaluated. Optimal rigid-body alignment can be computed efficiently using algorithms such as Kabsch alignment (Kabsch, 1976), but this optimisation must still be performed for every molecular comparison which makes a large-scale database search prohibitively expensive. Secondly, RMSD is only defined for molecules with identical atom counts as the minimisation requires a bijective correspondence between atoms which makes the cross-molecule comparison essential for virtual screening of structurally diverse databases undefined. Lastly, while RMSD itself is Lipschitz continuous under small coordinate perturbations, the optimal alignment may become unstable or non-unique for highly symmetric or near-degenerate configurations. This property makes RMSD unsuitable as a stability metric for conformers subject to thermal noise, as discussed earlier.

2.5 Efficient Similarity Search using K-d Trees

A core computational challenge of molecular similarity search is identifying conformers within a given distance threshold of a query conformer. A naive exhaustive approach requires $O(n^2)$ distance (Kurlin, 2024) computations which for PubChem's 943 million conformers, is computationally infeasible. K-d trees provide a spatial indexing structure that reduces this to $O(n \log n)$ construction time and $O(\log n)$ average query time for low-dimensional data.

A k-d tree is a binary space partitioning tree for k-dimensional data. At each internal node, the data is partitioned by a hyperplane perpendicular to one of the k coordinate axes with branches pruned during search when the minimum possible distance to any point in a subtree exceeds the query threshold. This process dramatically reduces distance computations compared to exhaustive search and is useful especially for large-scale data handling that would be necessary when processing the whole PubChem conformer dataset.

2.6 Force Fields and Conformer Generation

Force fields are mathematical models that approximate the potential energy of a molecule (Halgren, 1996). By treating atoms as classical particles rather than quantum mechanical wavefunctions, force fields allow stable 3D conformations to be found at a fraction of the computational cost of DFT. Specifically, PubChem uses the Merck Molecular Force Field 94 (MMFF94) which was developed by Halgren (1996) and selected for its good accuracy for small organic molecules and balance of speed and reliability across a wide range of chemical classes (Bolton et al., 2011). However, the principal limitation of MMFF94 is that it neglects explicit electron correlation effects which can be significant for molecules with unusual bonding. The GEOM dataset (Axelrod and Gómez-Bombarelli, 2022) addresses this by providing DFT-quality conformers at higher computational cost, though format obstacles prevented its use in this project as discussed in Section 2.2.2. The implications of using MMFF94 approximations rather than ground-truth experimental geometries are addressed in the construct validity discussion in Section 5.4.

2.7 Prior Work

The most directly related prior work is the MSc dissertation of Chatterjee (2025) conducted at the University of Liverpool under the same supervisors as this project. He applied SRD and SPD invariants to the full 1.3TB PubChem conformer dataset of 943 million conformers using the University of Liverpool's BARKLA HPC cluster. The primary contribution of Chatterjee's work was demonstrating the feasibility of invariant-based filtration at production scale by achieving search space reductions of 60–75% across atom count groups and identifying several interesting phenomena including duplicate conformer entries in PubChem and molecules with identical SPD vectors but different molecular formulae. A visualisation was provided as a static scatter plot generated using Plotly but exported as a non-interactive image showing SRD 1 against SRD diff for molecules with two to ten atoms. The plot provided a proof of concept that invariant-based molecular maps could reveal chemically meaningful structure because elongated molecules were observed to be forming a distinctive pattern and conformers of structurally similar molecules visibly clustered. This project extends Chatterjee's work by demonstrating that an invariant-based molecular map can also be interactive, empirically stable under perturbation and externally validated against RMSD by transforming a proof of concept into a research-ready tool.

3. Design

This section presents the design of the invariant-based molecular similarity search system. It was designed around five discrete components: a data pipeline; an invariant computation engine made up of three sub-components: the SRD module; the invariant engine responsible for SMILES and formula extraction and the SPD module; a filtration pipeline; an interactive visual map and a suite of validation experiments. Each component was designed with a specific research question in mind, and all the design decisions were deliberately made with alternatives considered where appropriate to ensure the system functions as intended. Figure 3.1 below demonstrates the overall architecture.

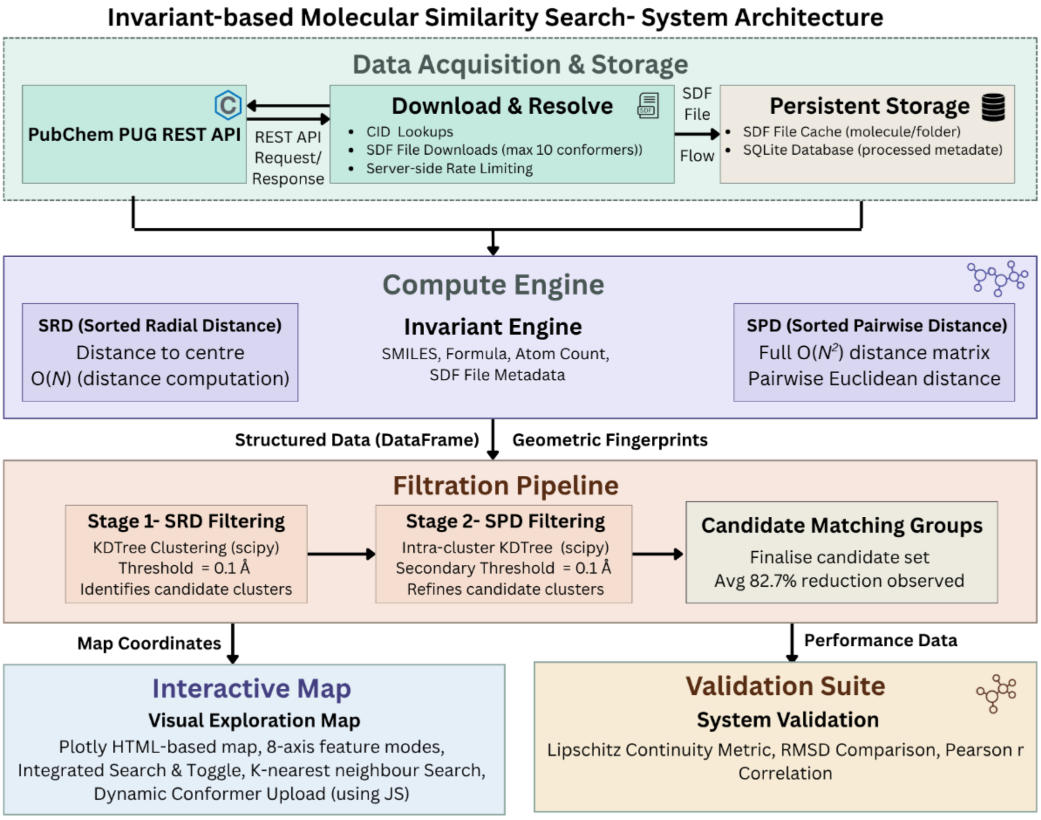


Figure 3.1- *Pipeline architecture showing the five system components, Data Acquisition & Storage + Compute Engine + Filtration Pipeline + Interactive Map and Validation Suite, and their data flow relationships.*

3.1 System Architecture

The system follows a linear pipeline architecture where each stage takes the output of the previous as its input, with the exception of the interactive map. The map visualises the full conformer dataset independently of the filtration results at the current scale, but in a production deployment, the filtration pipeline would serve as a pre-filter for similarity queries against the map to reduce the candidate set before full invariant comparison. The design was chosen over a modular microservice approach because the project operates on a fixed, pre-downloaded dataset rather than a live data stream so inter-service communication is an unnecessary overhead. As a key consideration, the pipeline was designed to be a single Python script so that the results would be easily verifiable by an external examiner or future researcher.

The five components are: the data acquisition and storage layer, responsible for fetching molecular structures from PubChem via its REST API and persisting them as SDF files; the invariant engine, which parses each SDF file and computes the Sorted Radial Distance (SRD) and Sorted Pairwise Distance (SPD) invariant vectors; the filtration pipeline, which uses a two-stage KDTree-based search to identify conformer groups to carry out a search space reduction; the interactive map, a standalone HTML visualisation that plots all conformers in invariant space with real-time search, plot, highlight and nearest-neighbour search; and lastly, the validation suite, which is comprised of two Lipschitz continuity experiments and RMSD comparison studies that give us empirical proof of the mathematical properties of the invariants.

3.2 Data Pipeline Design

The primary data source for this project is PubChem (Kim et al., 2023), the world's largest open repository of chemical information containing over 943 million conformers across approximately 116 million compounds (Chatterjee, 2025). The two access methods were considered were bulk FTP download and the PubChem REST API. The FTP approach would have provided faster raw throughput but required managing large, compressed archives and complex parsing logic so the REST API was selected instead. It allows targeted retrieval of specific compounds and conformers by name or CID which makes the dataset composition explicit and reproducible, but API access is slower and heavily dependent on load. On the other hand, it produces a transparent and well-documented dataset appropriate for study at the scale of the sample size.

The sample molecule list was curated to ensure diversity across chemical space so it covers nine unique categories: inorganic molecules, simple aliphatic organics, aromatic compounds, heteroaromatics, halogenated compounds, alcohols and phenols, amines, amino acids, sugars, nucleobases, vitamins, neurotransmitters and hormones, pharmaceuticals and natural products. A diverse sample is necessary to test whether the invariant-based map produces chemically meaningful groupings across structurally different molecule classes. In total, 1223 conformers across 250 unique molecules were downloaded from PubChem, with up to ten conformers fetched per compound where available. While modest in scale compared to the full PubChem conformer database of 943 million conformers, which Chatterjee (2025) processed using the University of Liverpool's BARKLA HPC cluster, this sample was sufficient to demonstrate the viability of the pipeline and for Lipschitz continuity experiments and an RMSD comparison. Scaling to the full dataset on the BARKLA HPC cluster is identified as future work.

Each compound was first resolved to its PubChem Compound Identifier (CID) via a name lookup, after which available conformer IDs were retrieved. For each conformer, the corresponding three-dimensional structure was downloaded as an SDF file at a rate limit of one request per 0.2 seconds to avoid overloading the PubChem servers during bulk download. To avoid redundant network requests, if the local molecules/ directory was non-empty, the download stage was skipped entirely. This design choice ensures that the resource and time-expensive network phase is only executed once and all subsequent stages operate on locally cached data.

3.3 Database Design

Column	Type	Description
id	INTEGER	Primary key, auto-increment
name	TEXT	Molecule name as downloaded from PubChem
cid	TEXT	PubChem compound ID extracted from SDF property block
formula	TEXT	Molecular formula computed by RDKit
smile	TEXT	SMILES string computed by RDKit
atom_count	INTEGER	Total atom count including hydrogen atoms

srd	TEXT (JSON)	Sorted radial distance vector, serialised as JSON array
spd	TEXT (JSON)	Sorted pairwise distance vector, serialised as JSON array

***Table 3.1**- Schema of the conformers table in the SQLite database, showing column names and data types.*

Once processed, conformer data is stored in an SQLite database. SQLite was selected over alternatives such as PostgreSQL or a flat CSV file because it requires no server process so the system can remain fully self-contained and portable. Secondly, it comes with structured query capabilities that would support extensions such as field-based filtering or nearest-neighbour queries from a backend service. Lastly, if the system were to be scaled to the full 943 million conformer dataset on the BARKLA HPC cluster. the SQLAlchemy ORM layer could replace SQLite for a production database with minimal refactoring required.

The database schema comprises a single conformers table with a sequential primary key, its molecule name, the PubChem CID, the conformer ID (a hexadecimal string made up of both the compound name and conformer index), the molecular formula, the SMILES string, the atom count and the SRD and SPD vectors serialised as JSON arrays. The SRD and SPD vectors were serialised rather than into a separate table because their length varies by molecule size so directly using the fixed-column relational schema is impractical. This step has another guard clause to ensure that the insert operation is skipped if the table is already populated which again, enforces idempotency.

3.4 Invariant Selection and Justification

The two geometric invariants selected for this project are Sorted Radial Distances (SRD) and Sorted Pairwise Distances (SPD). An invariant is a function of a molecular configuration that remains unchanged under isometry, so it remains the same under any combination of translation, rotation and reflection. This property is essential for molecular similarity search because any two representations of the same molecule in different orientations must produce identical descriptor vectors for the comparison to be meaningful (Anosova and Kurlin, 2025, Chapter 3). However, RMSD, the standard alignment-based metric, is not intrinsically invariant to rigid-body transformations and instead requires optimisation over rotations and translations to obtain the optimal superposition before comparison. In practice, this is typically solved using the Kabsch algorithm (Kabsch, 1976). This alignment step introduces computational disadvantage compared with descriptors that are invariant by construction.

Invariant	Time Complexity	Space Complexity	Completeness
SRD	O(n)	O(n)	Incomplete — can confuse some distinct molecules
SPD	O(n^2)	O(n^2)	Generically complete
RMSD	O(n^2) per pair	O(n)	N/A — alignment-based, not an invariant

Table 3.2- Complexity and completeness comparison of SRD, SPD and RMSD where n refers to atom count per molecule.

SRD is defined as the vector of Euclidean distances from each atom to the geometric centre of the molecule, sorted in descending order. It can be computed in O(n) time and O(n) space, so it is highly efficient. However, SRD is an incomplete invariant as two geometrically distinct molecules may in theory produce identical SRD vectors. Consider a square and a non-square rhombus with equal side lengths. They can have identical sorted radial distances despite being geometrically different. In practice this rarely occurs for real molecules, but the theoretical incompleteness makes the use of SPD a required filter.

SPD is defined as the vector of all pairwise Euclidean distances between atoms, sorted in ascending order. It requires O(n^2) time and space but is generically complete so for generic point configurations, it uniquely determines the molecular geometry up to Euclidean isometry (Crippen, 1988). The decision to sort in ascending order, placing the shortest pairwise distances first, was made on the recommendation of the project supervisor to reflect the chemical significance of short-range atomic interactions such as bond lengths. However, SRD is sorted in descending order because the largest radial distance (the atom furthest from the molecular centre) is the most informative feature when characterising overall molecular size.

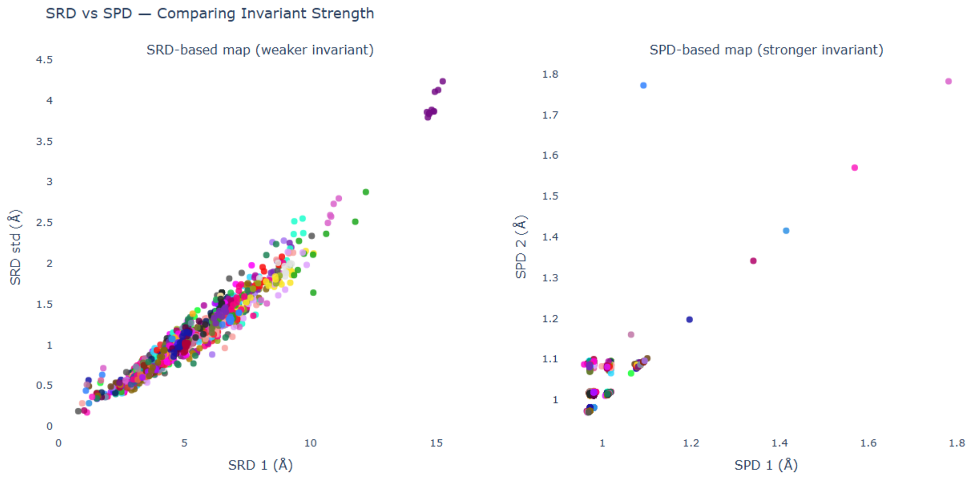


Figure 3.2 — SRD vs SPD invariant strength comparison. (a) SRD-based map (SRD 1 vs SRD std) showing conformers compressed into a single diagonal band, reflecting SRD's weaker discriminating power. (b) SPD-based map (SPD 1 vs SPD 2) showing greater separation between conformers, demonstrating SPD's stronger discriminating power as a generically complete invariant

3.5 Filtration Pipeline Design

The filtration pipeline was designed as a two-stage, coarse-to-fine search. In the first stage, conformers are grouped by atom count to compute SRD and SPD vectors. This is because they both have different lengths for molecules of different sizes, making cross-size comparison undefined. Within each atom-count group, a KDTree is constructed over the SRD matrix and queried for all pairs within a Euclidean distance threshold of 0.1 Å. This threshold was chosen to be below typical bond length variation but above floating-point noise to provide a meaningful similarity criterion.

Pairs identified in the SRD stage are passed to the second stage, where a KDTree is constructed over the SPD submatrix and queried with the same threshold. The two-stage design is motivated by computational efficiency because SRD filtering is O(n) and inexpensive to reduce the candidate set significantly before the more expensive SPD comparison is applied. This mirrors the design of database index structures, where a cheap primary index narrows the search before a more expensive secondary check is applied. A third stage where we take a direct coordinate comparison was identified as a design goal but not implemented as SPD filtering proved sufficient to distinguish conformer groups in the sample dataset.

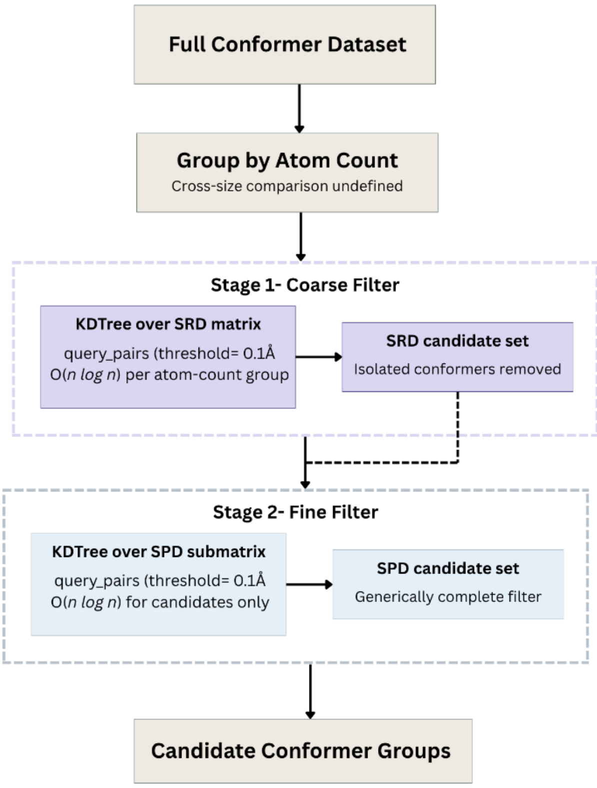


***Figure 3.3** — Two-stage coarse-to-fine filtration pipeline showing the flow from full conformer set to candidate groups.*

The use of scipy's KDTree implementation was preferred over a naive O(n^2) pairwise comparison for its O($n \log n$) construction time and O($\log n$) average query time, which becomes increasingly important as dataset size grows. At the scale of PubChem's full conformer database, an exhaustive pairwise comparison would be computationally infeasible, whereas a KDTree-based approach remains tractable.

The filtration pipeline is a standalone research contribution currently independent of the map. It provides empirical evidence for RQ4 by quantifying the search space reduction achievable through two-stage invariant-based filtering. Furthermore, it demonstrates the architectural foundation for a production-scale deployment processing PubChem's 943 million conformers. The filtration pipeline would serve as a pre-filter for similarity queries to reduce the candidate set before expensive SPD comparison. Therefore, the current implementation validates the approach at sample scale and scaling to production is identified as future work (Chatterjee, 2025).

3.6 Interactive Map Design

The interactive map was designed to allow researchers to visually explore the invariant space of the molecular dataset. The map plots two invariant-derived features directly with the default being SRD 1 (the largest radial distance, representing molecular size) on the x-axis and SRD standard deviation (representing shape spread) on the y-axis. *Together these two axes produce a map where molecules naturally separate by size and shape without any dimensionality reduction, with chemically similar molecules clustering visually in a manner consistent with chemical intuition*. The visualisation library Plotly was selected over alternatives such as Matplotlib which only produces static images unsuitable for interactive exploration and D3.js which would have required significant frontend development effort for no clear scientific benefit over Plotly, which exports a fully interactive, standalone HTML file requiring no server or runtime dependency. This is ideal because it makes the system easily shareable and suitable for submission as a project artefact.

Eight axis configurations were designed to expose different geometric properties of the molecules, covering molecular size, internal shape variation and short and long-range pairwise distances. The full set of axis options and their geometric interpretations are presented in Table 4.2. Each trace corresponds to a single molecule, and all conformers of that molecule share one trace so Plotly's native legend interaction (single click to hide, double click to isolate) can operate at a molecule level rather than the conformer level, which is the scientifically meaningful unit of analysis.

Three additional interactive features were designed: a text search panel that highlights traces by molecule name or molecular formula; a molecule upload panel that fetches a user-specified molecule from PubChem in real time, computes its SRD invariant in the browser, and plots it alongside the existing dataset; and a nearest-neighbour search panel that identifies the k most similar molecules in the 2D invariant projection space to a user-uploaded molecule. The nearest-neighbour search operates in SRD 1 vs SRD std space which is acknowledged as a limitation and full SPD vector nearest-neighbour search identified as future work.

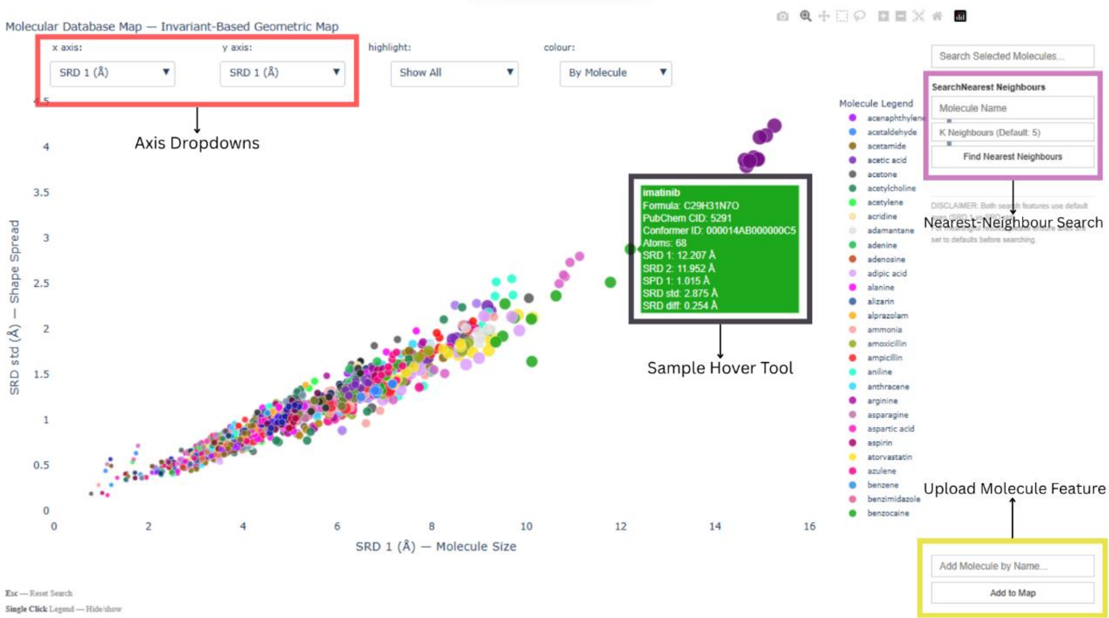


Figure 3.4 — *Design layout of the interactive molecular map with annotations for the axis dropdowns, search panel, upload panel, hover tooltip and nearest-neighbour panel.*

3.7 Validation Experiment Design

Two validation experiments were designed to provide empirical evidence for the mathematical properties of the invariants, addressing Research Questions 2 and 3 respectively.

3.7.1 Lipschitz Continuity Experiment

A Lipschitz-continuous function f satisfies the condition

$$||f(x) - f(y)|| \leq K||x - y||$$

for some constant K, meaning that small changes in input produce proportionally small changes in output. For a molecular invariant, this property guarantees stability under perturbation such as thermal noise where a molecule that vibrates slightly should map to a nearby point in invariant space, not an arbitrary distant one. This property is absent from RMSD, which can change discontinuously under small perturbations when the optimal alignment changes. The L-infinity norm was selected over the L2 norm because it captures worst-case deviation which is the most informative measure for stability analysis.

The experiment was designed as follows: for each of 50 molecules, the atomic coordinates were perturbed by a uniform random displacement in [−ε, ε] per atom per axis for seven values of ε: 0.001, 0.005, 0.01, 0.025, 0.05, 0.075, and 0.1 Å. Uniform noise was used here rather than Gaussian noise for simplicity and to ensure a well-defined perturbation bound. The SRD and SPD vectors were recomputed on the perturbed coordinates, and the change in each invariant was measured using the L-infinity norm (maximum absolute difference across all vector dimensions. Results were averaged across all 50 molecules per epsilon value. If the invariants are Lipschitz-continuous, the plot of epsilon against average invariant change should be approximately linear, with slope equal to the empirical Lipschitz constant K.

3.7.2 RMSD Comparison Experiment

This experiment was designed to validate SPD distance against the established standard technique, RMSD. Twenty-eight molecular pairs were selected from which, fourteen were of same-molecule pairs (two distinct conformers of the same compound) and fourteen different-molecule pairs that were further subdivided into structurally similar pairs (where low SPD distance was hypothesised) and structurally dissimilar pairs (where high SPD distance was hypothesised). This allows the analysis to test whether SPD distance correctly separates within-molecule variation from between-molecule variation.

For each pair, SPD distance was computed as the Root Mean Square (RMS) distance between the padded SPD vectors:

$$\frac{||SPD_1 - SPD_2||_2}{\sqrt{n}}$$

where n is the length of the longer vector. Normalisation by $\sqrt{n}$ was applied to make the distance accurately comparable across molecules of different sizes. RMSD was computed using RDKit's GetBestRMS function, which finds the optimal rigid-body alignment minimising RMSD. RMSD is only defined for molecules with identical atom counts so pairs with mismatched counts were recorded with RMSD undefined and retained for SPD distance analysis only.

Pearson correlation was selected as the primary measure of agreement between RMSD and SPD distance rather than alternatives such as Spearman rank correlation or mean absolute error because the hypothesis is specifically that RMSD and SPD distance have a linear relationship. Therefore, if SPD distance is a valid proxy for RMSD, the two should scale proportionally. Spearman rank correlation would only test whether the ordering is preserved, not whether the relationship is linear, which is the more informative claim. Mean absolute error was rejected because it is not scale-invariant and SPD distance and RMSD operate on different scales so direct error comparison is not possible without normalisation.

4. Implementation

This section describes the production of the system design presented in Section 3. The implementation was carried out in Python 3.11 using Visual Studio Code, with dependencies managed via a virtual environment. The key libraries used were RDKit for cheminformatics and SDF parsing; NumPy for numerical computation; pandas for data-frame management; scipy for the KDTree spatial index and Plotly for interactive visualisation. RDKit was chosen because it is the de facto standard for computational chemistry in Python; scipy's KDTree is a well-tested, production-quality spatial index and Plotly was the only library evaluated that could export a fully interactive standalone HTML file without a server dependency.

4.1 Data Fetching and SDF Parsing

The data fetching stage queries these three PubChem REST endpoints: a CID lookup endpoint, the conformer list endpoint and the SDF download endpoint. Asymmetric handling was necessary because PubChem's conformer endpoint returns HTTP 404 for molecules with no conformer diversity. Therefore, for molecules with multiple conformers available, up to ten conformers were downloaded with each being stored as a separate SDF file. For molecules with only the one conformer, a fallback to the standard 3D SDF endpoint was implemented.

SDF parsing was performed using RDKit's MolFromMolFile function with hydrogen retention enabled (removeHs=False). Retaining hydrogen atoms is essential for geometric accuracy because invariants computed on heavy-atom-only structures lose information about the true three-dimensional shape of the molecule, particularly for small molecules where hydrogen atoms constitute a significant fraction of the total atom count. The PubChem Compound CID and Conformer ID were extracted directly from the SDF property block by scanning the file text for the PUBCHEM_COMPOUND_CID and PUBCHEM_CONFORMER_ID tags. This approach was preferred over RDKit's GetPropsAsDict() method, which proved unreliable often returned null values when reading SDF files.

```
water.sdf
962
  -OEChem-05092609353D

  3  2  0     0  0  0  0  0  0999 V2000
    0.0000    0.0000    0.0000 O   0  0  0  0  0  0  0  0  0  0  0  0
    0.2774    0.8929    0.2544 H   0  0  0  0  0  0  0  0  0  0  0  0
    0.6068   -0.2383   -0.7169 H   0  0  0  0  0  0  0  0  0  0  0  0
  1  2  1  0  0  0  0
  1  3  1  0  0  0  0
M  END
> <PUBCHEM_COMPOUND_CID>
962

> <PUBCHEM_CONFORMER_RMSD>
0.4

> <PUBCHEM_CONFORMER_DIVERSEORDER>
1

> <PUBCHEM_MMFF94_PARTIAL_CHARGES>
3
1 -0.86
2 0.43
3 0.43

> <PUBCHEM_EFFECTIVE_ROTOR_COUNT>
0

> <PUBCHEM_PHARMACOPHORE_FEATURES>
2
1 1 acceptor
1 1 donor
```

Figure 4.1 - SDF file for water (H_2O, PubChem CID 962) showing the atom block with three-dimensional coordinates, bond block, and properties block containing PUBCHEM_COMPOUND_CID field extracted during parsing. PUBCHEM_CONFORMER_ID would be found much lower down in the properties block.

```
molecules
  acenaphthylene_000023C900000001.sdf
  acetaldehyde_000000B100000001.sdf
  acetamide_000000B200000001.sdf
  acetic_acid_000000B000000001.sdf
  acetone_000000B400000001.sdf
  acetylcholine_000000BB0000000A.sdf
  acetylcholine_000000BB0000000B.sdf
  acetylcholine_000000BB00000001.sdf
  acetylcholine_000000BB00000003.sdf
  acetylcholine_000000BB00000004.sdf
  acetylcholine_000000BB00000005.sdf
  acetylcholine_000000BB00000006.sdf
  acetylcholine_000000BB00000007.sdf
```

Figure 4.2 *— Local molecules directory containing SDF files for all downloaded conformers.*

4.2 SRD and SPD Implementation

The SRD computation begins by calculating the geometric centre of the molecule as the mean of all atomic coordinates. The Euclidean distance from each atom to this centre is then computed and the resulting vector is sorted in descending order (the atom furthest from the centre is at index zero). This is significant because the largest radial distance is the most size-discriminating feature for comparison and visualisation.

```python
def computeSRD(atoms):
    coords = np.array([[a['x'], a['y'], a['z']] for a in atoms])
    center = coords.mean(axis=0)
    distances = np.sqrt(((coords - center) ** 2).sum(axis=1))
    return sorted(distances, reverse=True)
```

Figure 4.3 *— Python implementation of the SRD invariant computation.*

The SPD computation begins by iterating over all unique atom pairs using a nested loop and computing the Euclidean distance for each pair. The resulting distances are sorted in ascending order so the shortest pairwise distance is at index zero. This ascending sort order was specified by the project supervisor to reflect the chemical significance of short-range interactions which represent tightest bonds between atoms. The $O(n^2)$ complexity of SPD computation is acceptable for molecules in the dataset (maximum of 96 atoms, giving at most 4,560 pairwise distances per conformer in the dataset) but would require optimisation using vectorised NumPy operations or some other method before deployment at the scale of the full PubChem dataset.

```python
def computeSPD(atoms):
    coords = np.array([[a['x'], a['y'], a['z']] for a in atoms])
    distances = []
    n = len(coords)
    for i in range(n):
        for j in range(i+1, n):
            d = np.sqrt(((coords[i] - coords[j]) ** 2).sum())
            distances.append(d)
    return sorted(distances)
```

Figure 4.4 *— Python implementation of the SPD invariant computation.*

A worked example for validation: for water (H_2O, three atoms), the SRD vector has three elements corresponding to the distances from each of the two hydrogen atoms and the oxygen atom to the molecular centre. The SPD vector has three elements corresponding to the three unique pairwise distances: O-H_1, O-H_2, and H_1-H_2. Manual computation of the SRD vector from the PubChem conformer coordinates (CID 962) yields [0.7890, 0.7890, 0.3977] Å, with the two hydrogen atoms equidistant from the geometric centre confirming the near-symmetric geometry of water. This matches the code output of [0.7890, 0.7890, 0.3978] Å rounded to four decimal places. Note that the discrepancy of 0.0001 Å in the third element is a result of floating-point arithmetic and is below any physically meaningful threshold. The full worked calculation is provided in Appendix C

4.3 KDTree Filtration

The filtration pipeline groups conformers by atom count and constructs a scipy KDTree over the SRD for each group. The query pairs method is called with a threshold of 0.1 Å to identify all pairs within that distance. Any conformer that appears in at least one such pair is added to a candidate set for secondary filtering in the SPD phase. Conformers that do not appear in any pair are geometrically distinct from all others in the group and are excluded from further consideration, as they represent unique structures that do not require disambiguation, consistent with the functionality of Chatterjee's (2025) KDTree. This graph-based grouping logic where we find all non-isolated vertices in the KDTree proximity graph, was preferred over connected-component clustering because the goal is to identify candidates for further inspection, not to produce a definitive partition of the conformer space.

Atom Count	Total Conformers	After SRD	After SPD	Reduction
10	12	2	2	83.3%
14	65	14	10	84.6%
16	33	6	4	87.9%
17	43	4	4	90.7%
20	78	8	8	89.7%
29	32	2	2	93.8%
33	40	2	2	95.0%
24	49	49	49	0.0%
25	20	20	20	0.0%
40	21	21	21	0.0%

Table 4.1 — Representative filtration results showing atom count, total conformers, conformers after SRD filter, conformers after SPD filter and percentage reduction. The sample represents atom count groups with high reduction and zero reduction. Full results are presented in Appendix A.

Search space reduction is defined as the percentage decrease in the number of conformers requiring pairwise comparison within each atom count group. Contextually, it refers to the removal of geometrically isolated conformers with no similar neighbours within the 0.1 Å threshold so only candidate groups of near-duplicate or highly similar conformers for further analysis are left. This reduces the number of conformers requiring detailed pairwise comparison.". The two distinct patterns visible in the filtration table above are groups where the pipeline achieves substantial reduction and groups where reduction is zero. The zero-reduction rows are not a failure of the pipeline but indicate atom count groups where no conformers were found within the 0.1 Å threshold of each other, meaning all conformers in those groups are geometrically distinct and correctly retained as unique structures.

4.4 Interactive Map

The interactive map is implemented as a Plotly go.Figure with one trace per molecule. This design decision reflects Plotly's legend interaction operating at the trace level, meaning that a single click on a legend entry hides or shows all conformers of that molecule simultaneously, which is the scientifically meaningful unit of analysis. If each conformer were a separate trace, the legend would contain hundreds of entries, making it unusable. Marker size was scaled proportionally to atom count within each trace for additional visuals of molecular size that complements the SRD 1 x-axis without requiring a separate legend.

Axis switching is implemented via Plotly's updatemenus mechanism, which injects pre-computed data arrays into the figure at render time. Eight dropdown configurations were built using a makebuttons helper function, each replacing the x or y data arrays of all traces simultaneously. A bug was encountered during development in which both the x and y dropdowns were updating only the x-axis data which left the y-axis unchanged regardless of selection. This was traced to an incorrect use of Plotly's update method arguments where the layout update dictionary was being passed as a second data update rather than as a separate layout argument. The fix required separating the data update (a list of dictionaries, one per trace) from the layout update (a single dictionary with axis title keys) into the correct positional arguments of the args list.

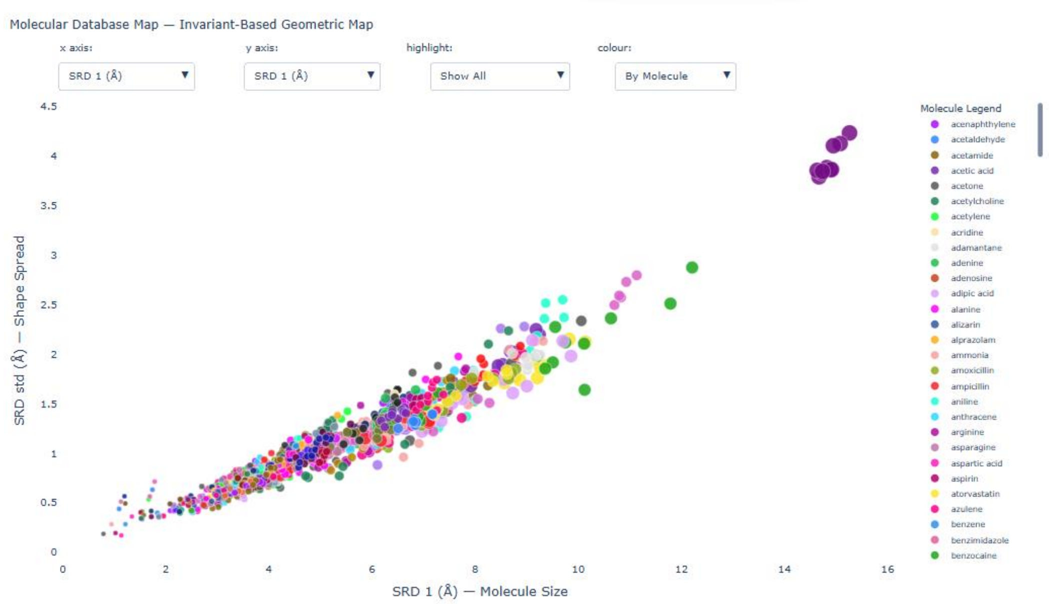


Figure 4.5 *— Interactive molecular map showing conformers in SRD 1 vs SRD std space, coloured by molecule.*

Three colour modes were implemented via a separate dropdown: colour by molecule (default, one colour per unique molecule name); colour by formula type (grouping molecules by elemental composition into eight categories such as C+H only, C+H+O and halogenated compounds) and colour by atom count (a continuous blue-to-red scale proportional to atom count). These modes were implemented as Plotly restyle calls rather than separate traces, avoiding the performance overhead of redrawing the figure.

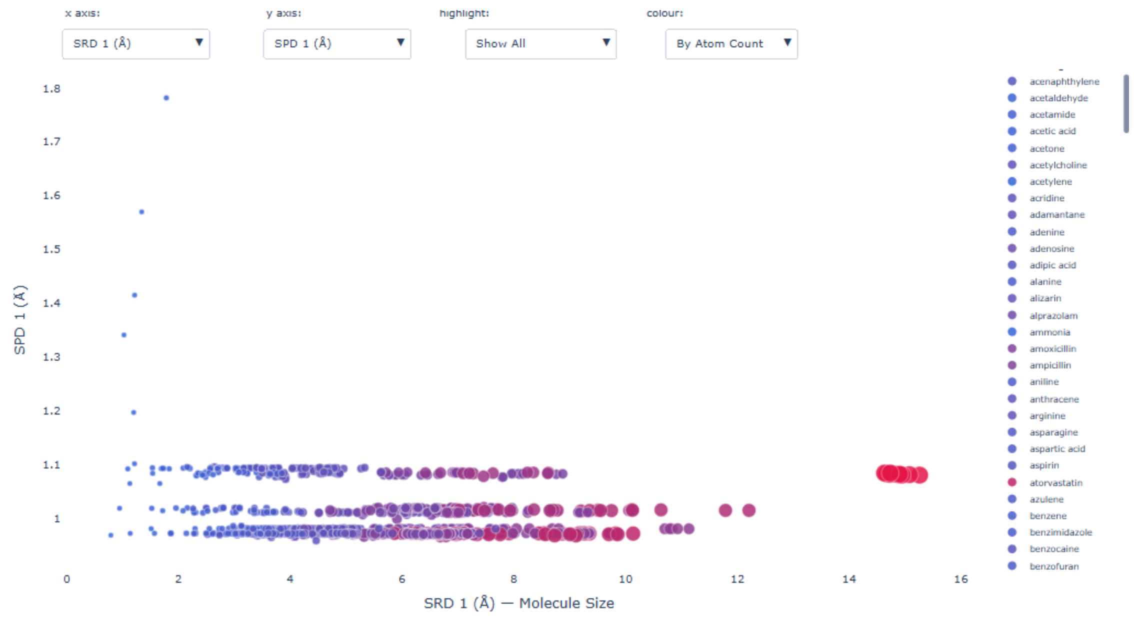


Figure 4.6 *- Interactive map with SRD 1 (molecular size) on the x-axis and SPD 1 (shortest pairwise distance) on the y-axis. The narrow horizontal bands between 0.9–1.1 Å show the near-constant nature of bond lengths across chemically diverse molecules separated by molecules size. This view demonstrates that different axis configurations expose distinct geometric properties of the invariant space.*

The interactive map includes eight axis configurations that are each derived directly from the SRD or SPD invariant vectors and each encoding a distinct geometric property of the molecular structure. Unlike dimensionality reduction techniques, these axes retain their geometric meaning so a value on the x-axis corresponds to a specific, interpretable property of the molecule. Table 4.2 presents the full set of axis options with their geometric interpretations. The default axes are SRD 1 on the x-axis and SRD standard deviation on the y-axis and were selected because they jointly capture the two very visually discriminating properties across a chemically diverse dataset. The remaining six axes allow a researcher to interrogate the dataset from alternative geometric perspectives such as short-range atomic bonds via SPD 1 and the gap between the two outermost atoms via SRD diff, without any loss of mathematical interpretability. Figure 4.6 above illustrates the SPD 1 axis configuration which corresponds to bond lengths and varies little across chemically diverse molecules which is why a distinctive horizontal band is produced, contrasting the wide variation in molecular size captured by SRD 1.

Axis	Definition	Geometric significance
SRD 1	Largest radial distance from geometric centre	Overall molecular size
SRD 2	Second largest radial distance	Secondary size measure
SRD 3	Third largest radial distance	Tertiary size measure
SRD mean	Mean of all radial distances	Average atomic spread

SRD std	Standard deviation of all radial distances	Shape symmetry — low for symmetric molecules, high for asymmetric
SRD diff	SRD 1 – SRD 2	Gap between two outermost atoms — near zero for linear molecules
SPD 1	Shortest pairwise atomic distance	Shortest bond length approximation
SPD 2	Second shortest pairwise atomic distance	Secondary short-range interaction

Table 4.2- Summarises the eight axis configurations available in the interactive map and their geometric interpretation

4.5 Browser Side Features

Three interactive features were added to the exported HTML file by injecting JavaScript directly into the document body after Plotly had rendered. This injection approach was taken because Plotly's Python API does not expose hooks for custom JavaScript so the HTML export provides a completely self-contained document that can be extended post-generation. The features that were added are as follows:

The molecule search panel listens to keypress events on a text input and dynamically adjusts the opacity of all traces based on whether the trace name or formula matches the query string. Matching traces are set to opacity 0.95 and non-matching traces to 0.04 to produce a visual highlighting effect without removing any data from the plot.

The molecule upload panel fetches a user-specified molecule from PubChem's REST API directly from the browser, parses the returned SDF text to extract atom coordinates, recomputes the SRD invariant in JavaScript and finally adds the molecule as a new trace. The user-added molecule is explicitly highlighted and labelled. This feature demonstrates that the invariant computation is sufficiently simple to be reimplemented client-side in approximately thirty lines of JavaScript which reflects the O(n) simplicity of the SRD algorithm.

Finally, the nearest-neighbour search panel fetches the query molecule from PubChem, computes its SRD 1 and SRD standard deviation and performs a brute-force O(n) scan over all existing traces to identify the k nearest neighbours in the 2D invariant projection space. The axes are reset to the default configuration (SRD 1 vs SRD std) before the search runs to ensure that the distance computation and the visual display are consistent. A disclaimer is displayed in the UI noting that the search operates in the default 2D space and that axis-aware nearest-neighbour search is identified as future work because computing axes configurations calculated using the SPD invariant are far more computationally expensive. Results are deduplicated by molecule name so that multiple conformers of the same molecule do not occupy multiple result slots.

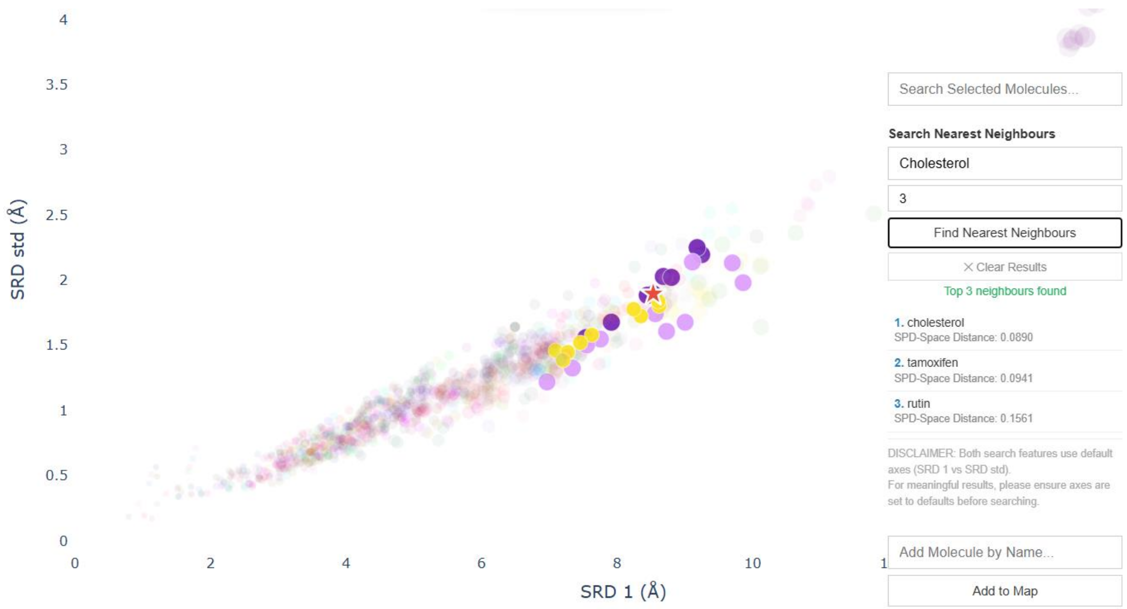


Figure 4.7 *— Nearest-neighbour search result for cholesterol (k=3), showing the query molecule as a red star and the three nearest conformers highlighted.*

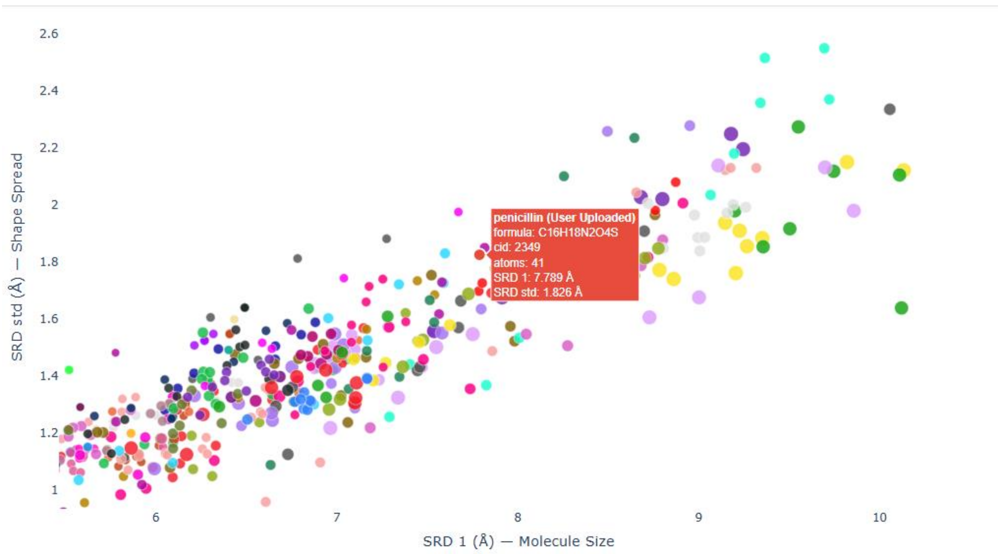


Figure 4.8 *— User-uploaded molecule plotted on the map alongside existing dataset conformers.*

4.6 Lipschitz Validation

The Lipschitz validation experiment is the primary novel contribution of this project.

```python
for r in results[:50]:
    original_srd = np.array(r['srd'])
    original_spd = np.array(r['spd'])

    for eps in epsilons:
        coords = np.array([[a['x'], a['y'], a['z']] for a in r['atoms']])

        # Perturbation is applied
        perturbation = np.random.uniform(-eps, eps, coords.shape)
        perturbed_coords = coords + perturbation

        # SRD is recomputed on perturbed coords.
        center = perturbed_coords.mean(axis=0)
        perturbed_srd = np.array(sorted( ···

        # recompute SPD on perturbed coords
        n = len(perturbed_coords)
        perturbed_spd_list = []
        for i in range(n): ···
        perturbed_spd = np.array(sorted(perturbed_spd_list))

        # The change is measured using L-infinity Norm
        srd_change = np.max(np.abs(perturbed_srd - original_srd))
        spd_change = np.max(np.abs(perturbed_spd - original_spd))

        perturbation_results.append({
            'molecule': r['name'].rsplit(' ', 1)[0],
            'epsilon': eps,
            'srd_change': srd_change,
            'spd_change': spd_change,
            'atom_count': r['atom_count']
        })
```

Figure 4.9 *– Lipschitz Continuity experiment implementation in VS Code. Atomic coordinates are perturbed by uniform noise in [−ε, ε] and the change is measured using the L-infinity norm. The collapsed sections correspond to the SPD and SRD recalculation code.*

For each of the first 50 molecules in the dataset, the atomic coordinates were perturbed by a uniform random displacement drawn from $[-\varepsilon, \varepsilon]$ per atom per axis for each of seven epsilon values. The random seed was fixed to 42 before each perturbation to ensure reproducibility. Then, SRD and SPD were recomputed on the perturbed coordinates and the change in each invariant was measured using the L-infinity norm. This process was repeated for all 50 molecules and all 7 epsilon values, giving 350 data points per invariant.

Two visualisations were produced. The first is a line graph plotting the mean invariant change against epsilon for both SRD and SPD. The second is an on-map visualisation showing the displacement of six selected molecules under ε=0.1 Å perturbation. Both are presented and analysed in Section 5.2.

4.7 RMSD Comparison

For the RMSD comparison, each molecule in a pair was converted from the internal atom list representation to an RDKit RWMol object using a helper function that iterates over the internal atom list, adds each atom by its element symbol and assigns three-dimensional coordinates directly to a Conformer object. This approach bypasses RDKit's standard SDF parsing pipeline, which was necessary because the internal atom list representation used throughout the pipeline does not correspond to a file on disk.

RDKit's GetBestRMS function was then called, which internally solves the optimisation problem discussed in 3.4 to find the rotation minimising RMSD before computing the final value. SPD distance was computed as described in Section 3.7.2. Pairs with mismatched atom counts were recorded with RMSD set to None and excluded from the Pearson correlation but their SPD distances were retained for qualitative analysis. Of the 28 pairs, only 11 produced valid RMSD values and the remaining 17 pairs involved molecules of different atom counts.

These 14 different-molecule pairs were deliberately chosen to include structurally dissimilar molecules of varying sizes, so atom count mismatches were inevitable. By design, these pairs were retained for SPD distance analysis since SPD distance handles molecules of different sizes through zero-padding of the shorter vector. Finally, SPD distance was computed as the normalised L2 norm. Unlike RMSD, this requires no molecular alignment and operates directly on the sorted invariant vectors. The normalisation by $\sqrt{n}$ ensures distances are comparable across molecule pairs of different sizes so larger molecules would not systematically produce larger raw distances due to having more pairwise distances in their SPD vector.

Figure 4.10 *— Python function converting the internal atom list representation to an RDKit molecule object for RMSD computation.*

```python
def atoms_to_rdmol(atoms):
    mol = RWMol()
    conf = Conformer(len(atoms))
    for i, atom in enumerate(atoms):
        mol.AddAtom(Chem.Atom(atom['symbol']))
        conf.SetAtomPosition(i, Point3D(atom['x'], atom['y'], atom['z']))
    mol = mol.GetMol()
    mol.AddConformer(conf, assignId=True)
    return mol
```

4.8 Problems Encountered

Several non-trivial problems were encountered during the implementation stage. Each is described below:

PubChem API timeouts. Early on, Intermittent HTTP 503 errors occurred during bulk SDF download, particularly for large molecules with many conformers. These were handled by wrapping each request in a try-except block and logging failures, allowing the pipeline to continue with partial data rather than terminating.

GEOM dataset format obstacles. The GEOM dataset (Axelrod and Gomez-Bombarelli, 2022) provides significantly more accurate DFT-quality conformers over PubChem's MMFF94 conformers. It was identified as a secondary data source in the original project proposal, but GEOM distributes conformer data in a custom compressed JSON format that proved incompatible with RDKit's standard SDF parsing pipeline. An attempt was made to write a custom parser, but the format's schema was insufficiently documented to complete this within the project timeline. GEOM processing was therefore descoped and is acknowledged as a limitation.

Windows encoding issues with the Å symbol. Terminal output containing the Ångström symbol (Å) caused UnicodeEncodeError on Windows when redirected to a file as Windows PowerShell defaults to CP1252 encoding rather than UTF-8. This was resolved by setting the PYTHONIOENCODING environment variable to utf-8 before execution.

Axis dropdown bug. The x and y axis dropdowns both updated the x-axis data, so the user was unable to configure the y axis. The root cause was that Plotly's updatemenus args parameter expects a list containing a dictionary of trace data updates and a dictionary of layout updates in that order. The layout update dictionary was incorrectly being passed as a second data update dictionary, which Plotly silently ignored. The fix was to restructure the args list to pass the layout update as a separate second element.

Plotly typed array incompatibility with nearest-neighbour search. The nearest-neighbour search initially returned zero results despite correctly fetching and computing the query molecule's invariants. Debugging revealed that after Plotly renders a figure, trace coordinate arrays are stored internally as typed binary objects rather than plain JavaScript arrays (specifically as: {dtype: 'f8', bdata: '...', _inputArray: Float64Array}), meaning that standard Array.from(trace.x) returns an empty array rather than the coordinate values. The fix required accessing the underlying Float64Array directly via trace.x._inputArray before converting to a plain array. This is an undocumented internal behaviour of Plotly's binary data optimisation and was identified through Chrome's inspect tool console.

RDKit SDF property parsing. RDKit's GetPropsAsDict() method did not reliably extract the PUBCHEM_COMPOUND_CID and PUBCHEM_CONFORMER_ID fields from downloaded SDF files. This was resolved by reading the SDF file as plain text and scanning line-by-line for those property tags and extracting the value from the following line. This approach is more robust because it does not depend on RDKit's internal property parser which was not compatible with certain SDF metadata formats.

5 Testing and Evaluation

This Section presents the testing and evaluation of the invariant-based molecular similarity map. Testing covers unit verification of the core algorithmic components and evaluation assesses the system as a whole against the four research questions defined in Section 1.2. Each research question is addressed by presenting the relevant evidence, interpreting the results clearly and acknowledging limitations. It concludes with an analysis of equivalence relation properties and a threats to validity assessment.

5.1 Unit Testing

Unit testing was performed on each major algorithmic component of the pipeline to verify their correctness before integration. Tests were conducted manually rather than using an automated testing framework such as pytest because the pipeline is implemented as a single sequential script rather than a modular package. An automated test suite is unnecessary at the current scale but is identified as a priority for future development if the pipeline expanded to the full PubChem conformer dataset to enable regression testing.

5.1.1 SPD Verification

The SPD invariant was verified for water (H_2O) by manually computing the three unique pairwise distances between the oxygen and two hydrogen atoms. Using the coordinates from the PubChem SDF file (CID 962) (see Appendix C):

$$d(O, H^1) = \sqrt{(0.0000 - 0.2774)^2 + (0.0000 - 0.8929)^2 + (0.0000 - 0.2544)^2} = \sqrt{0.0770 + 0.7973 + 0.0647} = \sqrt{0.9390} = \mathbf{0.9690}\,\text{Å}$$

$$d(O, H^2) = \sqrt{(0.0000 - 0.6068)^2 + \left(0.0000 - (-0.2383)\right)^2 + \left(0.0000 - (-0.7169)\right)^2} = \sqrt{0.3682 + 0.0568 + 0.5139} = \sqrt{0.9389} = \mathbf{0.9690}\,\text{Å}$$

$$d(H^1, H^2) = \sqrt{(0.2774 - 0.6068)^2 + \left(0.8929 - (-0.2383)\right)^2 + \left(0.2544 - (-0.7169)\right)^2} = \sqrt{0.1085 + 1.2795 + 0.9435} = \sqrt{2.3315} = \mathbf{1.5270}\,\text{Å}$$

Sorting in ascending order gives $SPD(H^2O) = [0.9690, 0.9690, 1.5270]\text{Å}$ which corresponds with the two O–H bond lengths followed by the H–H distance. The computeSPD function returned $[0.9690, 0.9690, 1.5270]\text{Å}$, confirming the correctness of the implementation. The equal O–H distances reflect the symmetric geometry of the water molecule. The SPD verification was performed for a three-atom molecule only because the $O(n^2)$ number of pairwise distances makes manual verification impractical. Automated property-based testing such as, verifying that SPD is invariant under random rotations and translations for a sample of molecules is identified as future work.

5.1.2 Axis Dropdown Verification

The axis dropdown bug described in Section 4.8 was discovered during unit testing and resolved by setting the same axis option on both the x and y dropdowns. If both dropdowns correctly update their respective axes, selecting the same axis for both should produce a perfect diagonal scatter plot because every conformer plotted is at (x, x) since the x and y values are identical. Figure 5.1 shows the result of selecting SRD 1 on both axes. This visual test confirms that the layout update dictionary is correctly separated from the data update dictionary in the Plotly updatemenus args list.

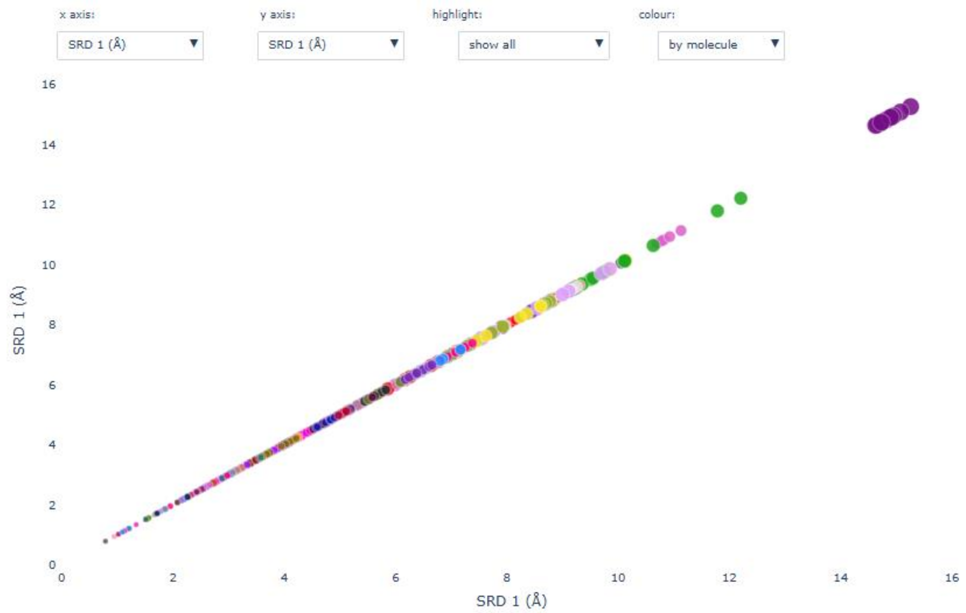


__Figure 5.1__ — Axis dropdown verification. Selecting SRD 1 on both x and y axes produces a perfect diagonal.

5.1.3 Conformer Clustering Verification

A conformer clustering sanity check was performed by uploading paracetamol (acetaminophen) via the browser-based molecule upload feature. Since paracetamol is present in the sample dataset with multiple downloaded conformers, a correctly implemented upload feature should place the newly fetched structure in the same region of invariant space as the existing dots. Figure 5.2 shows the uploaded paracetamol (red circle) landing adjacent to the existing paracetamol cluster which confirms that the browser-side SRD computation and PubChem API integration are functioning correctly.

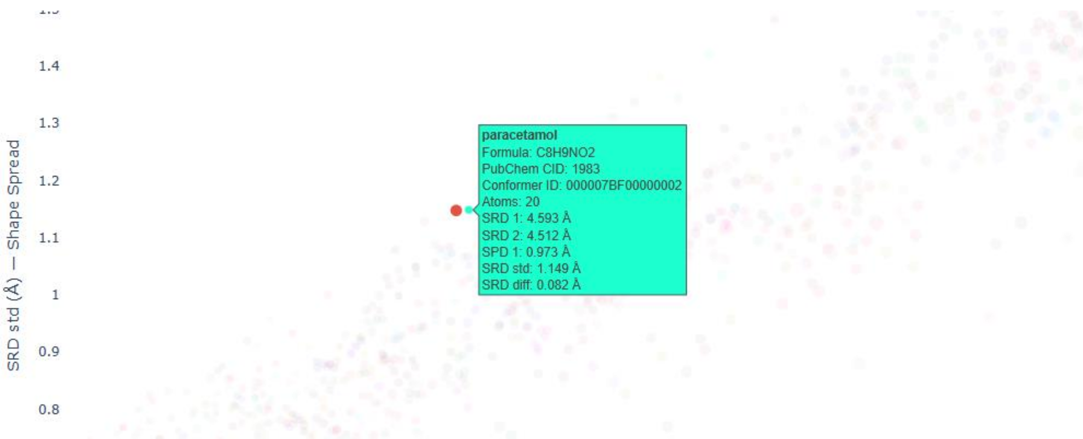


__Figure 5.2__ — Conformer clustering sanity check. The uploaded paracetamol molecule (red circle, fetched live from PubChem) lands in close proximity to an existing paracetamol conformer (blue circle) in the dataset which confirms correct browser-side invariant computation.

5.2 Evaluation Against Research Questions

Each of the four research questions defined in Section 1.2 derived from the project proposal is addressed in turn, following the structure: statement of the RQ, presentation of evidence, interpretation, conclusion on whether the RQ is answered and acknowledgement of limitations:

5.2.1 RQ1 Interactive Molecular Map

RQ1: Can sorted distance invariants (SRD/SPD) produce an interactive molecular map that preserves chemical intuitions about molecular similarity?

The interactive molecular map was successfully implemented as a standalone Plotly HTML file plotting 1,223 conformers across approximately 250 unique molecules in invariant space. The map provides eight axis configurations, three colour modes, a molecule search panel, a molecule upload panel and a nearest-neighbour search feature. Figure 5.3 shows the full map in its default configuration (SRD 1 vs SRD std, coloured by molecule).

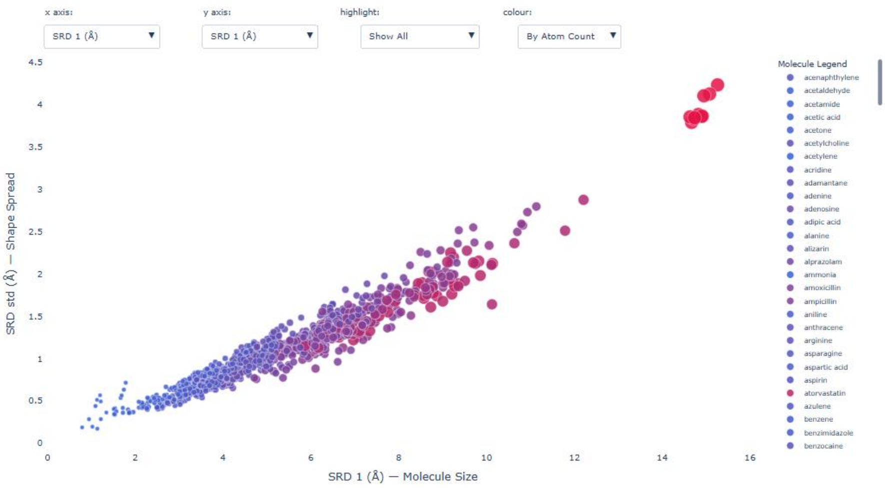


Figure 5.3 *— Interactive molecular map configured so each point represents one conformer, coloured by atom count. Larger molecules are coloured in red with smaller molecules in blue. A clear gradient can be seen, placing the largest molecules with a high SRD 1 and high SRD std in the top right. Marker size is also proportional to atom count*

The map preserves several chemical facts that support its validity as a molecular similarity representation within chemical descriptor space (Lipkus and Yuan, 2009). Amino acids cluster in a distinct mid-range region of the map, consistent with their similar molecular sizes and three-dimensional geometries. Secondly, large drug molecules such as cholesterol and erythromycin occupy the far right of the SRD 1 axis. Small inorganic molecules such as water and ammonia are confined to the lower left and aromatic compounds form a recognisable band at intermediate SRD 1 values with low SRD standard deviation to reflect their compact, symmetric ring structures. These groupings emerge directly from the geometric invariants without any dimensionality reduction.

Figure 5.4 shows the colour-by-formula-type mode, in which molecules are coloured by elemental composition. The separation of chemical families visible in this view provides further qualitative evidence that the invariant space encodes chemically meaningful structure.

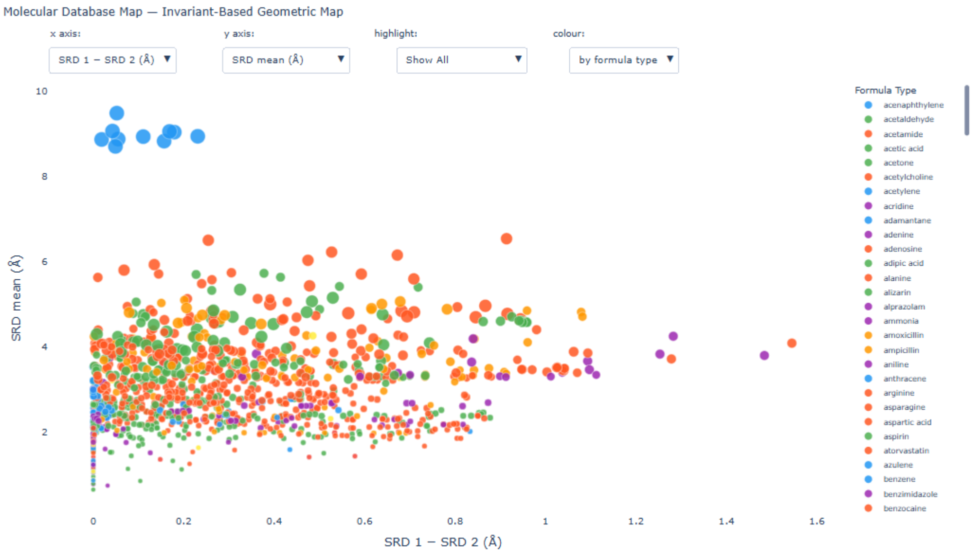


Figure 5.4 *— Interactive map coloured by formula type conformers, plotting SRD diff (x-axis) against SRD mean (y-axis). The large blue marker at upper left is beta carotene ($C_{40}H_{56}$), a large symmetric hydrocarbon correctly identified as C+H only. The colour mode can be toggled between colouring by molecule, formula type and atom count without redrawing the figure in real-time. Formula type categories search largest to smallest combinations from: blue = C+H only, green = C+H+O, purple = C+H+N, orange = C+H+N+O, amber = C+H+N+O+S, yellow = C+H+S, red = halogen, grey = other. Marker size is proportional to atom count. Marker size is proportional to atom count.*

Qualitative evidence was also gathered from informal feedback. Informal feedback from the project supervisor indicated that the system met the research standard expected for the project. While this does not constitute a formal user evaluation, it provides meaningful qualitative evidence from a domain expert that the system produces outputs consistent with chemical intuition.

In conclusion, the invariant-based molecular map successfully produces a visually interpretable representation of chemical space that preserves chemical facts about molecular similarity. The clustering of chemical families and the correct positioning of molecules by size and symmetry support the validity of the approach. However, no formal user study with computational chemists was conducted so feedback was informal and limited to two individuals. Therefore, a structured user study evaluating the map's utility is identified as future work.

5.2.2 RQ2 - Lipschitz Continuity

RQ2: Are SRD and SPD Lipschitz-continuous under small atomic perturbations, and if so, what is the empirical Lipschitz constant?

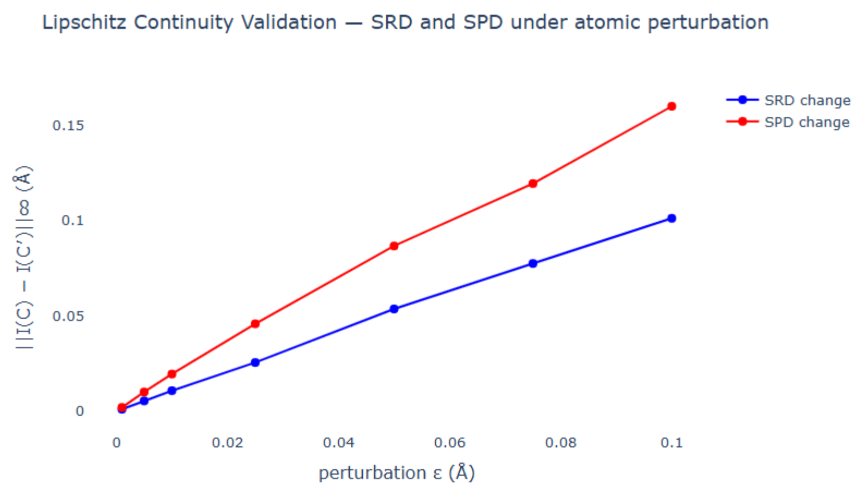
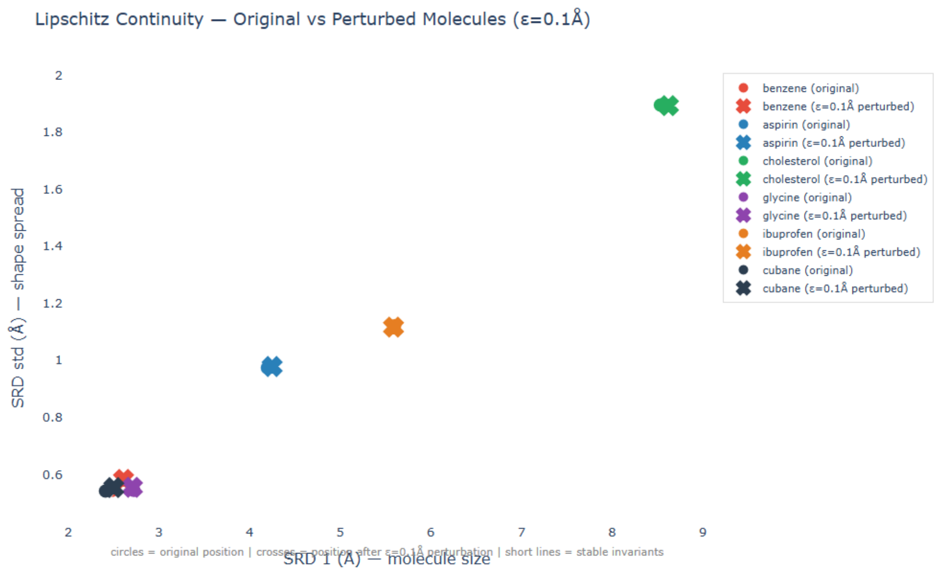


***Figure 5.5** — Lipschitz continuity validation. Mean invariant change (L-infinity norm) as a function of perturbation magnitude ε for SRD (blue) and SPD (red). Both invariants demonstrate an approximately linear relationship, consistent with Lipschitz continuity.*

The Lipschitz continuity experiment described in Section 3.7.1 was conducted across 50 molecules and seven perturbation magnitudes ($\varepsilon \in \{0.001, 0.005, 0.01, 0.025, 0.05, 0.075, 0.1\}$Å). Figure 5.5 shows the mean invariant change as a function of ε for both SRD and SPD.

Both SRD and SPD demonstrate an approximately linear relationship between ε and mean invariant change, consistent with the Lipschitz condition $||I(C) - I(C')|| \leq K||C - C'||_{\infty}$. The empirical Lipschitz constant K was estimated by fitting a linear model to the epsilon vs mean invariant change data using numpy.polyfit and K is taken as the slope of the fitted line. When calculated, gives $K \approx 1.045$ for SRD and $K \approx 1.592$ for SPD. SPD exhibits a steeper slope which possibly indicates greater sensitivity to atomic perturbation, which is expected because SPD's stronger discriminating power encodes more geometric information and is therefore more responsive to structural changes.



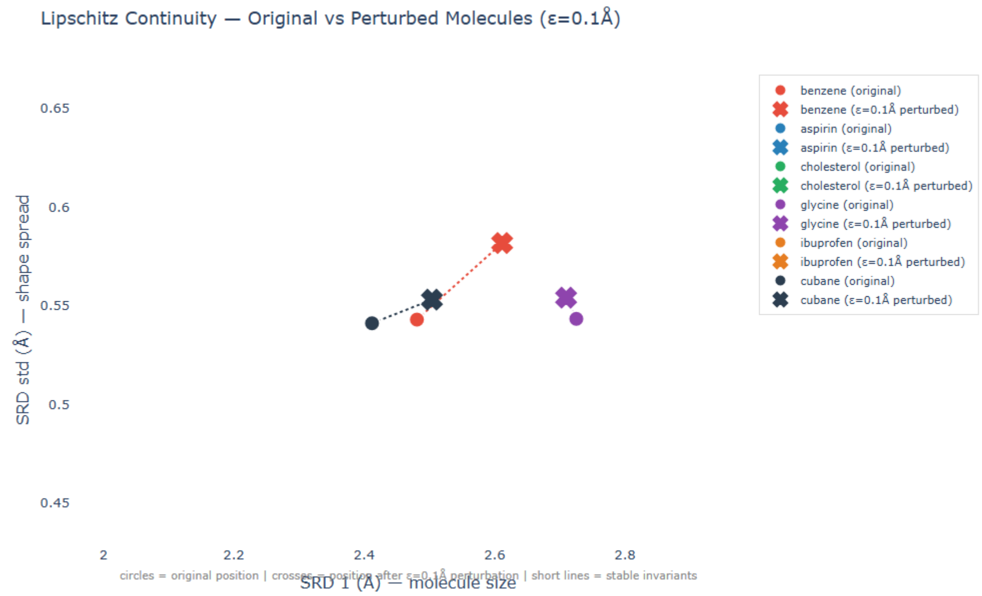


***Figure 5.6** — On-map Lipschitz perturbation visualisation. (a) Full map view showing all conformers with original positions (circles) and perturbed positions after ε=0.1 Å perturbation (crosses). (b) Zoomed view showing benzene, cubane and glycine with short connecting which confirms small, bounded displacement in invariant space consistent with Lipschitz continuity.*

Figure 5.6 shows the on-map Lipschitz visualisation for six selected molecules, displaying original positions (circles) and perturbed positions after ε = 0.1 Å perturbation (crosses), connected by dotted lines. Rigid, symmetric molecules such as benzene and cubane displace less in invariant space than large, flexible molecules such as ibuprofen and cholesterol. This is consistent with chemical intuition where a rigid molecule's geometry changes little under small perturbations, but a flexible molecule's conformational variation is more pronounced. This result provides additional qualitative support for the validity of the invariants as stability-preserving descriptors.

Therefore, RQ2 is answered empirically. Both SRD and SPD exhibit approximately linear growth in invariant change as a function of perturbation magnitude, consistent with Lipschitz continuity. The empirical Lipschitz constants are estimated from the slope of the fitted lines. However, this is empirical validation and not a formal mathematical proof of Lipschitz continuity. A rigorous proof would require bounding the worst-case change analytically, which is beyond the scope of this feasibility study. The perturbation used uniform noise but real thermal vibrations follow a different distribution. Results may therefore overstate stability under physically realistic noise and is further addressed further in Section 5.4.

5.2.3 RQ3 – SPD Distance and RMSD

RQ3: How well does SPD distance correlate with RMSD for same-molecule conformer pairs versus different-molecule pairs?

The RMSD comparison experiment was conducted across 28 molecular pairs with 14 same-molecule conformer pairs and 14 different-molecule pairs (further subdivided into structurally similar and dissimilar pairs as described in Section 3.7.2). Of the 28 pairs, 11 produced valid RMSD values and the remaining 17 pairs involved molecules of different atom counts for which RMSD is undefined. Figure 5.7 presents the scatter plot of RMSD against SPD distance for all 11 valid pairs.

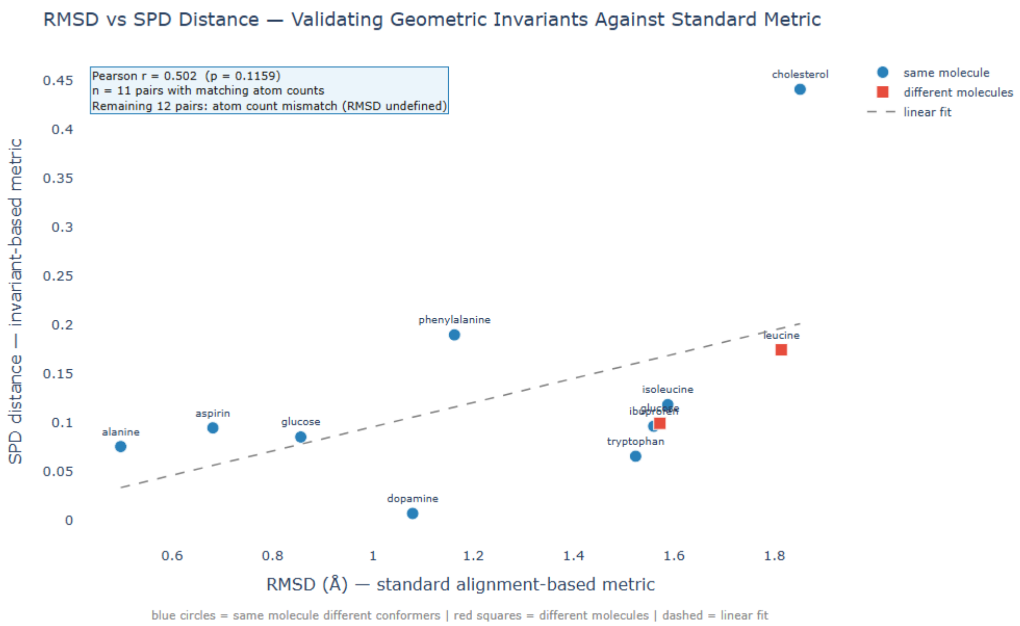


Figure 5.7 *— RMSD vs SPD distance scatter plot. Blue circles represent same-molecule conformer pairs; red squares represent different-molecule pairs. The dashed line shows the linear fit across all valid pairs.*

A moderate positive correlation was observed between RMSD and SPD distance ($Pearson\ r\ =\ 0.502, p\ =\ 0.116, n\ =\ 11$). The positive direction of the correlation is consistent with the hypothesis that SPD distance and RMSD measure related aspects of structural similarity, so pairs that are geometrically dissimilar by RMSD tend to have higher SPD distances. However, the correlation did not reach statistical significance at the conventional $p < 0.05$ threshold, which is attributable to the small sample of atom-count-matched pairs rather than an absence of the underlying relationship.

The largest deviation from the expected correlation was the dopamine same-molecule conformer pair which produced a notably low SPD distance (0.007) despite a moderate RMSD (1.07 Å). This is consistent with the known property of the invariants that SRD and SPD are rotation and translation invariant, whereas RMSD requires alignment and is sensitive to the choice of optimal rotation. For a flexible molecule such as dopamine, two conformers may differ significantly in their spatial orientation, and produce a high RMSD while their invariant vectors remain similar and produce a low SPD distance. This illustrates both a strength and a limitation of the invariant approach because it correctly identifies structural similarity regardless of orientation but may underestimate geometric diversity for highly flexible molecules.

The SPD distances alone also tell a coherent story. Extremely structurally similar pairs (glucose vs fructose: 0.099; leucine vs isoleucine: 0.174) produced substantially lower SPD distances than structurally dissimilar pairs (benzene vs cholesterol: 7.005; water vs aspirin: 4.263), which is consistent with chemical facts about the relative structural similarity of these molecule pairs. These results suggest that SPD distance, despite requiring no molecular alignment and operating in $O(n^2)$ rather than $O(n^3)$ per pair, captures similar structural relationships to RMSD and is a computationally efficient proxy for molecular similarity.

Therefore, RQ3 is partially answered because moderate positive correlation ($r = 0.502$) was observed between RMSD and SPD distance. Although it supports the hypothesis that SPD distance is a valid proxy for structural similarity, statistical significance was not achieved due to the small sample of 11 atom-count-matched pairs. Even though the sample of 28 pairs was designed to be balanced across similarity categories, many had mismatched atom counts which reduced the valid sample. Extending the analysis to 50–100 pairs with controlled atom count matching is identified as a priority for future work.

5.2.4 RQ4 – Filtration Search Space Reduction

RQ4: How effectively does the two-stage SRD to SPD filtration reduce the conformer search space?

The filtration pipeline was applied to the full sample dataset of 1,223 conformers across 57 unique atom count groups. Table 5.1 presents a representative selection of results, but the complete table is provided in Appendix A.

Atom Count	Total Conformers	After SRD	After SPD	Reduction
10	12	2	2	83.3%
14	65	14	10	84.6%
16	33	6	4	87.9%
17	43	4	4	90.7%
20	78	8	8	89.7%
29	32	2	2	93.8%
33	40	2	2	95.0%
24	49	49	49	0.0%
25	20	20	20	0.0%
40	21	21	21	0.0%

Table 5.1 — Representative filtration results. Full results are presented in Appendix A.

The filtration results are reported here as a standalone evaluation of search space reduction and integration of the filtration pipeline as a pre-filter for nearest-neighbour queries (currently performs a brute-force scan of all plotted conformers) against the interactive map is identified as a natural extension of this work. Across the 19 atom count groups where the pipeline identified at least one conformer pair within the 0.1 Å similarity threshold, the average search space reduction was 82.7% with a maximum reduction of 95.0% for the 33-atom group. This is significant because a naive exhaustive pairwise comparison over 1,223 conformers would require $1{,}223 \times 1{,}222 / 2 = \mathbf{747{,}253}$ comparisons. An 82.7% reduction in the candidate set reduces this to approximately 129,000 comparisons and saves of over 600,000 redundant comparisons on the sample set. At the scale of PubChem's 943 million conformers (Chatterjee, 2025), the saving would be proportionally far larger.

The 38 atom count groups that showed 0% reduction are not a failure of the pipeline but indicates groups where no conformers fell within the 0.1 Å similarity threshold, meaning all conformers in those groups are geometrically distinct and are correctly retained as unique structures. This is the expected behaviour for a diverse, curated sample dataset: most molecules were deliberately chosen to be chemically different, so few conformers within the same atom count group should be geometrically similar.

Therefore, RQ4 is answered because the two-stage filtration pipeline successfully achieves an average reduction of 82.7% and a maximum of 95.0% across groups where similar conformers exist. These results show that an invariant-based approach is viable as a search space reduction strategy. The two-stage design also demonstrates its computational efficiency because atom count groups SRD filtering reduced the candidate set before the more expensive O(n^2) SPD was applied. Because results are based on a 1,223-conformer sample dataset curated for chemical diversity, there is high rate of 0% reduction groups but in a real-world database search over chemically similar molecules, the filtration pipeline would likely produce higher average reductions. Scaling to the full PubChem dataset on the University of Liverpool's BARKLA HPC cluster is identified as the primary future work.

5.3 Equivalence Relation Properties

A key theoretical requirement of the system is that the similarity relation defined by the invariants satisfies the axioms of an equivalence relation: reflexivity, symmetry and transitivity. Therefore, this section assesses the extent to which these properties hold. The term d(C, C') denotes the SPD distance between conformers C and C which is computed as the normalised L2 norm between their padded SPD vectors as defined in Section 3.7.2.

Reflexivity ensures that $d(C, C) = 0$. It is guaranteed by construction because for any conformer C, SRD(C) = SRD(C) and SPD(C) = SPD(C), so the distance between a conformer and itself is zero regardless of the invariant used or the threshold applied. This axiom requires no empirical verification (Anosova and Kurlin, 2025, Chapter 1).

Symmetry ensures that $d(C, C') = d(C', C)$ and is also guaranteed by construction. The SPD distance is computed as the L2 norm between two padded vectors, which is symmetric by definition. Similarly, KDTree query_pairs returns unordered pairs and the threshold comparison is symmetric. Therefore, no asymmetric distance can arise from this implementation (Anosova and Kurlin, 2025, Chapter 1).

Transitivity ensures that if $C \sim C'$ and $C' \sim C''$ $then$ $C \sim C''$ and is the most complex of the three properties and cannot be guaranteed in general for threshold-based similarity relations. It is important to distinguish between transitivity of the distance metric which does hold because the L2 norm satisfies the triangle inequality, and transitivity of the equivalence relation defined by the threshold. For a threshold ε, it is possible that $d(C, C') \leq \varepsilon$ and $d(C', C'') \leq$

ε but $d(C, C'') > \varepsilon$ which would be a transitivity violation (Anosova and Kurlin, 2025, Chapter 1). Transitivity of the equivalence relation was empirically assessed through the conformer clustering results. Conformers of the same molecule, which should be transitively related as members of the same equivalence class, were consistently grouped together by the pipeline and there were no observed cases where two conformers of the same molecule were placed in separate groups via different intermediate conformers. Therefore, the pipeline does demonstrate empirical transitivity for the sample dataset but just does not formally prove it. Formal transitivity testing sampling 10,000 triplets and measuring the proportion of violations at thresholds $\varepsilon \in \{0.01, 0.05, 0.1\}$ Å is identified as future work. The supervisor confirmed that SPD is generically complete (V. Kurlin, personal communication, 2025) which provides theoretical grounding for the lack of transitivity violations in practice.

5.4 Threats to Validity

This section identifies and discusses threats to the validity of the experimental results, following the standard framework of internal, external, construct and reliability validity. Critically interrogating the limitations of the results is essential when fitting findings accurately within the broader research context.

5.4.1 Internal Validity

Internal validity covers whether the experimental results accurately reflect what they claim to measure and are free from confounding factors.

The Lipschitz continuity experiment uses uniform random noise in $[-\varepsilon, \varepsilon]$ as the perturbation model. Real molecular thermal vibrations do not follow a uniform distribution and are more accurately modelled by a Maxwell-Boltzmann distribution weighted by atomic mass and temperature. The use of uniform noise may therefore overstate the stability of the invariants under physically realistic perturbations so future work should repeat the experiment using a physically motivated noise model to produce more realistic Lipschitz estimates. Although the fixed random seed (42) used throughout the Lipschitz experiment ensures reproducibility, the results reflect a single realisation of the random perturbation whereas different seeds would produce slightly different empirical Lipschitz constants. To quantify this sensitivity, the experiment should be repeated across multiple seeds and the variance in K reported. This would strengthen the claim that the observed linearity is a robust property of the invariants rather than an artefact of a particular random sample.

5.4.2 External Validity

External validity concerns whether the results generalise beyond the specific sample used in this study.

The sample dataset of 1,223 conformers across approximately 250 unique molecules was deliberately curated for chemical diversity rather than proportional representation of PubChem's chemical space. This decision was made to stress-test the invariants across structurally varied molecules rather than to mirror the actual distribution of molecule types in PubChem, which is heavily weighted toward drug-like organic compounds. The high rate of 0%

reduction groups in the filtration results further reflects this diversity because a production deployment searching over chemically similar molecules would likely produce higher average reductions. The filtration results therefore represent a conservative lower bound on the pipeline's effectiveness in practice.

It is also important to note that the MMFF94 conformers available from PubChem represent computational approximations rather than experimentally determined ground-truth geometries. PubChem's MMFF94 conformer generation rules also restrict conformers to molecules with at most 50 non-hydrogen atoms and at most 15 rotatable bonds. Larger, more flexible molecules common in drug discovery are not represented in the sample. Whether the invariants remain discriminating for molecules beyond this size range is an open question.

5.4.3 Construct Validity

Construct validity concerns whether the measurements used in the study accurately capture the theoretical constructs they are intended to measure.

The RMSD comparison experiment uses RMSD as the ground truth measure of structural similarity. However, RMSD and SPD measure related but fundamentally distinct properties with RMSD measuring the minimum achievable coordinate deviation after optimal rigid-body alignment and SPD distance measuring the difference in the sorted set of all pairwise interatomic distances. For example, the dopamine conformer pair discussed in Section 5.2.3 which has a high RMSD but low SPD distance is not an error but a genuine case where the two metrics disagree because SPD is alignment-free and RMSD is not. The Pearson correlation between the two metrics could therefore be interpreted as a measure of their agreement, not as a measure of the correctness of either. Similarly, the empirical Lipschitz constant K estimated from the perturbation experiment is a property of the specific molecules tested and the specific noise distribution used, so is not a universal property of SRD or SPD and should not be extrapolated beyond the conditions of this experiment.

5.4.4 Reliability

Reliability concerns whether the results are reproducible under the same conditions.

The fixed random seed of 42 within the Lipschitz experiment ensures it is fully reproducible and running the script again on the same dataset will produce identical results. Similarly, the filtration pipeline is also deterministic given the same input data and threshold as well as the interactive map which is generated from the same pipeline output so will be identical across runs.

A possible exception is the browser-based nearest-neighbour search which relies on live PubChem API calls so is dependent on network availability and PubChem server response times. If PubChem returns a slightly different SDF structure for the same molecule (for example, due to a database update), the query molecule's invariants and therefore its nearest neighbours may differ across sessions. This is a minor reliability concern for the current implementation though and would be resolved by caching query results locally if necessary, in a future version.

6. Project Ethics

This project was conducted in accordance with the University of Liverpool's ethical guidelines and falls under Data Category A and Participant Category 0 because there was no personal data collected or processed and no human participants were involved in the research

6.1 Data Sources

All molecular data used in this project was sourced exclusively from PubChem (Kim et al., 2023) which a publicly available database maintained by the National Institutes of Health (NIH) of the United States. PubChem requires no registration for access and makes all compound and conformer data freely available under an open data policy. Therefore, no proprietary, confidential or personally identifiable data was used at any point in the project. The molecules included in the sample dataset are publicly known compounds spanning established chemical categories which are all documented in the open scientific literature. The project did not collect, store or process any data relating to individuals. The nearest-neighbour search feature in the interactive map makes live API calls to PubChem from the user's browser, but no data from these interactions is stored or logged by the system. The system does not require user authentication and does not collect any usage data.

6.2 Human Participants

No formal study involving human participants was conducted and only informal feedback was received from the project supervisor during the project presentation and from a second marker who expressed interest in using the tool for research. This feedback was not part of a structured evaluation protocol so no consent forms were required, no data was recorded and no individuals can be identified from the feedback described in Section 5.2.1.

6.3 Dual-Use Considerations

The same technology that aids drug discovery could theoretically be applied to the design of harmful compounds. However, the system does not generate novel molecular structures and only classifies and compares existing conformers from a public database. Furthermore, the system provides no information beyond what is already publicly available through PubChem's own web interface. The project therefore presents no meaningful increase in dual-use risk beyond what is already present in existing public tools.

7. Conclusion and Future Work

This section summarises the contributions of the project against each research question, reflects on what did not go to plan and identifies directions for future work.

7.1 Summary

The project addressed four research questions that covered whether sorted distance invariants could produce an interactive molecular map preserving chemical intuitions about similarity (RQ1). The interactive map was successfully implemented as a standalone Plotly HTML file plotting 1,223 conformers across approximately 250 unique molecules in invariant space, with eight configurable axis options, three colour modes, molecule search, an upload feature and nearest-neighbour search. Chemical families cluster naturally in the invariant space with amino acids occupying a distinct mid-range region, aromatic compounds forming a compact band at intermediate SRD 1 values and large drug molecules such as cholesterol sitting at the far right of the size axis. This demonstrates that the invariant axes encode genuine chemical meaning without dimensionality reduction. Therefore, RQ1 is answered.

RQ2 asked whether SRD and SPD are Lipschitz-continuous under small atomic perturbations. The empirical Lipschitz continuity experiment was conducted across 50 molecules and seven perturbation magnitudes and confirmed an approximately linear relationship between perturbation magnitude and invariant change for both SRD ($K \approx 1.045$) and SPD ($K \approx 1.592$). Therefore, RQ2 is answered empirically with the caveat that formal mathematical proof remains future work.

RQ3 asked how well SPD distance correlates with RMSD for same-molecule versus different-molecule pairs. A moderate positive correlation was observed (Pearson $r = 0.502$, n = 11 atom-count-matched pairs) which consistent with the initial hypothesis that SPD distance and RMSD measure related aspects of structural similarity. Statistical significance was not achieved at the $p < 0.05$ threshold possibly due to the small sample of atom-count-matched pairs and extending this analysis is identified as future work. Therefore, RQ3 is partially answered.

RQ4 asked how effectively the two-stage filtration pipeline reduces the conformer search space. Across 19 atom count groups containing similar conformers, the pipeline achieved an average reduction of 82.7% with a maximum of 95.0% so the invariant-based approach meaningfully reduces the search space compared to exhaustive comparison. Therefore, RQ4 is answered.

Beyond the research questions, the project delivered several additional contributions including a browser-based molecule upload feature that computes invariants client-side in JavaScript and a nearest-neighbour search feature operating in invariant projection space. Additionally, a single reproducible Python script produces all outputs which lowers the barrier to adoption compared to Chatterjee's (2025) multi-script HPC pipeline.

7.2 What Did Not Go to Plan

The GEOM dataset (Axelrod and Gómez-Bombarelli, 2022) which uses DFT-quality conformers compared to PubChem's MMFF94 structures, could not be processed due to format obstacles. This was anticipated as a risk in the original proposal.

GFlowNet integration was an optional aim in the original proposal and was descoped entirely as the core aims were sufficiently ambitious and the supervisor confirmed that the invariant-based map and validation experiments were the priority contributions. UMAP and t-SNE dimensionality reduction were dropped on the supervisor's recommendation because these methods are stochastic and produce axes with no geometric meaning and therefore, contradict the motivation for using mathematically grounded invariants.

7.3 Future Work

The most scientifically significant direction for future work that directly motivates scaling the pipeline to production is to map empty regions of invariant space. Plotting the full PubChem conformer dataset and identifying regions of invariant space where no known molecules sit could reveal opportunities for novel molecular design because molecules occupying empty regions may represent synthesisable novel structures or provide additional insights on the range of conformers that exist on the full dataset. However, processing all 943 million conformers would require additional computational resources such as the University of Liverpool's BARKLA HPC cluster. Chatterjee (2025) demonstrated the feasibility of this at the processing stage so integrating the interactive map with a much larger subset is a natural next step.

A potential route for future work includes expanding to the GEOM dataset (Axelrod and Gómez-Bombarelli, 2022) and comparing DFT-quality invariant maps against MMFF94-based maps to quantify the error introduced by force field approximations. This was a construct validity concern identified in Section 5.4. Another is extending the RMSD comparison from 11 valid pairs to 50–100 atom-count-matched pairs for a more statistically significant external validation of SPD distance for structural similarity. Furthermore, future work could also include sampling triplets from the conformer dataset and measuring the proportion of transitivity violations to formally quantify the equivalence relation properties of the pipeline or implementing a full nearest-neighbour search, which currently operates only in the default SRD 1 vs SRD std projection. However, this would require conformer data to be available in the browser or via a backend API which is a major undertaking. Lastly, a structured evaluation with computational chemists to assess the map's utility for virtual screening tasks using established usability metrics would provide new qualitative and quantitative evidence for research question 1.

8 BCS Criteria and Self-Reflection

8.1 BCS Project Criteria

The below table checks the project against the BCS criteria:

Criterion	How this project meets the criterion
Practical and Analytical Skills	The project required Python programming (RDKit, pandas, scipy, plotly, numpy packages), JavaScript and HTML for browser-side features, SQLite database design, REST API integration and the application of geometric mathematics including Euclidean distance computation, linear algebra for invariant vectors and statistical analysis using Pearson correlation. Analytical skills were demonstrated through the design of the validation suite including a Lipschitz continuity experiment and the RMSD comparison study and the threats to validity assessment.
Innovation and Creativity	The interactive invariant-based molecular map with live PubChem integration, nearest-neighbour search and browser-side invariant computation for user specified molecules represents a novel tool that extends the state of the art beyond Chatterjee (2025). The Lipschitz continuity validation is a novel empirical measurement of this property for SRD and SPD. The browser-side reimplementation of the SRD algorithm in JavaScript demonstrates creative application of the theoretical foundations to a practical user-facing feature.
Synthesis of Information	The project integrates knowledge from computational chemistry (molecular conformers, force fields, SMILES), geometric mathematics (isometry invariants, Lipschitz continuity), software engineering (pipeline

	design, efficient data fetching, database design), data visualisation (Plotly, interactive HTML) and statistics (Pearson correlation, threats to validity). This synthesis across chemistry, mathematics, software engineering and data science reflects the highly interdisciplinary nature of the project.
Real-world Need	Drug discovery relies on molecular similarity search at scale so this project addresses a genuine computational bottleneck since PubChem's 943 million conformers cannot be searched exhaustively, and existing alignment-based methods are computationally expensive and mathematically unstable. The system developed in this project provides a working prototype for alignment-free, similarity search with empirically validated stability properties. The supervisor's offer of a research continuation opportunity indicates that the tool has potential for real-world scientific impact.
Self-management	The project was completed independently across two semesters and required some scope changes (UMAP dropped, GEOM descoped), technical obstacles (SDF property parsing, Plotly binary array format) and deadline pressures for this project and other modules. The project proposal, presentation and dissertation were all delivered on schedule.
Critical Self-evaluation	Limitations are acknowledged throughout the dissertation including the 1,223-conformer sample size, the absence of an automated test suite, the statistical insignificance of the RMSD correlation and the restrictions of the browser-based nearest-neighbour search. The threats to validity section (Section 5.4) is a formal framework for interrogating the results and reflects an honest assessment of what the project achieved and what remains to be done.

Table 8.1 – Project contributions against the BCS assessment criteria.

8.2 Self-Reflection

The interdisciplinary nature of this project made it one of the most technically demanding experiences of my degree. Almost every component in this project was a combination of new concepts in computational chemistry, statistics or geometric mathematics so early on, terms like conformer, isometry, Lipschitz continuity and MMFF94 force field were unfamiliar and intimidating. However, by the end, I can discuss the generic completeness of SPD invariants in productive technical conversations. That learning arc is something I am genuinely proud of, and it reflects the value of taking on a project at the boundary of multiple disciplines rather than staying within a comfortable single domain.

In terms of what went well, the interactive map exceeded my initial expectations because the decision to implement browser-side invariant computation in JavaScript for live molecule upload and nearest-neighbour search was a significant engineering challenge that I had not anticipated at the proposal stage. The Lipschitz validation experiment also produced cleaner results than expected and the near-unity empirical Lipschitz constant for SRD ($K \approx 1.045$) is an elegant result that shows the invariant changes by almost exactly the perturbation magnitude.

The sequential nature of the pipeline meant that debugging required re-running the entire script from scratch whenever an issue was identified. Therefore, if repeated, I would recommend starting with a Jupyter notebook rather than a monolithic script so individual cells could be run in isolation. It would have reduced the most time during the debugging of the Plotly binary array issue (Section 4.8) and the SDF property parsing problem. Nevertheless, the supervisor's offer of a research continuation opportunity following the project presentation was the most meaningful external validation of the project's value and genuinely confirmed that the work produced something useful to the research community, rather than just a completed module assignment and is motivation for the future work directions identified in Section 7.3.

APPENDICES

Appendix A - Complete Filtration Results

Table A.1 presents the complete filtration results across all 58 atom count groups in the sample dataset of 1,223 conformers.

Atoms	Total	After SRD	After SPD	Reduction
3	5	5	5	0.0%
4	5	5	5	0.0%
5	5	5	5	0.0%
6	3	3	3	0.0%
7	2	2	2	0.0%
8	8	8	8	0.0%
9	7	7	7	0.0%
10	12	2	2	83.3%
11	5	5	5	0.0%
12	12	12	12	0.0%
13	11	2	2	81.8%
14	65	14	10	84.6%
15	23	4	4	82.6%
16	33	6	4	87.9%
17	43	4	4	90.7%
18	16	2	2	87.5%
19	42	20	20	52.4%
20	78	8	8	89.7%
21	14	2	2	85.7%
22	30	6	6	80.0%
23	51	6	6	88.2%
24	49	49	49	0.0%
25	20	20	20	0.0%
26	55	55	55	0.0%
27	26	2	2	92.3%
29	32	2	2	93.8%
30	15	4	4	73.3%
31	39	39	39	0.0%
32	56	12	12	78.6%
33	40	2	2	95.0%
35	21	21	21	0.0%

36	1	1	1	0.0%
37	30	30	30	0.0%
38	1	1	1	0.0%
39	17	2	2	88.2%
40	21	21	21	0.0%
41	10	10	10	0.0%
42	5	5	5	0.0%
43	41	41	41	0.0%
44	31	31	31	0.0%
45	27	27	27	0.0%
47	30	30	30	0.0%
48	23	23	23	0.0%
49	21	21	21	0.0%
51	20	20	20	0.0%
53	2	2	2	0.0%
54	12	12	12	0.0%
55	10	10	10	0.0%
56	9	4	4	55.6%
57	11	11	11	0.0%
63	10	10	10	0.0%
65	10	10	10	0.0%
68	20	20	20	0.0%
73	10	10	10	0.0%
74	8	8	8	0.0%
76	10	10	10	0.0%
96	10	10	10	0.0%

Table A.1 – Full filtration table results post-K-d filtering.

Appendix B – Full Molecule List

Molecule	CID	**Molecule**	CID	**Molecule**	CID
acenaphthylene	9161	**ethanol**	702	**phenanthrene**	995
acetaldehyde	177	**ethyl acetate**	8857	**phenazine**	4757
acetamide	178	**ethylamine**	6341	**phenol**	996
acetic acid	176	**ethylbenzene**	7500	**phenylalanine**	6140
acetone	180	**ethylene**	6325	**phosphine**	24404
acetylcholine	187	**ethylene glycol**	174	**phthalic anhydride**	6811
acetylene	6326	**fluorene**	6853	**piperidine**	8082

Molecule	CID	**Molecule**	CID	**Molecule**	CID
acridine	9215	**fluorobenzene**	10008	**piperine**	638024
adamantane	9238	**fluoroethane**	9620	**porphyrin**	66868
adenine	190	**fluoxetine**	3386	**prednisolone**	5755
adenosine	60961	**folic acid**	135398658	**procaine**	4914
adipic acid	196	**formaldehyde**	712	**progesterone**	5994
alanine	5950	**formamide**	713	**proline**	145742
alizarin	6293	**formic acid**	284	**propane**	6334
alprazolam	2118	**fructose**	2723872	**propanol**	1031
ammonia	222	**furan**	8029	**propionic acid**	1032
amoxicillin	33613	**gaba**	119	**propylene**	8252
ampicillin	6249	**galactose**	6036	**pseudoephedrine**	7028
aniline	6115	**gefitinib**	123631	**purine**	1044
anthracene	8418	**geraniol**	637566	**pyrazine**	9261
arginine	6322	**glucosamine**	439213	**pyrene**	31423
asparagine	6267	**glucose**	5793	**pyridazine**	9259
aspartic acid	5960	**glucuronic acid**	94715	**pyridine**	1049
aspirin	2244	**glutamate**	33032	**pyridoxine**	1054
atorvastatin	60823	**glutamic acid**	33032	**pyrimidine**	9260
azulene	9231	**glutamine**	5961	**pyrrole**	8027
benzene	241	**glycerol**	753	**pyrrolidine**	31268
benzimidazole	5798	**glycine**	750	**pyrroline**	94395
benzocaine	2337	**guanine**	135398634	**quercetin**	5280343

Molecule	CID	Molecule	CID	Molecule	CID
benzofuran	9223	**guanosine**	135398635	**quinazoline**	9210
benzothiophene	7221	**heptane**	8900	**quinine**	3034034
benzyl alcohol	244	**hexane**	8058	**quinoline**	7047
benzylamine	7504	**histamine**	774	**quinoxaline**	7045
beta carotene	5280489	**histidine**	6274	**ranitidine**	3001055
biotin	171548	**hydrocortisone**	5754	**resorcinol**	5054
bromobenzene	7961	**hydrogen cyanide**	768	**resveratrol**	445154
bromoethane	6332	**hydrogen peroxide**	784	**retinol**	445354
butane	7843	**hydrogen sulfide**	402	**riboflavin**	493570
butanol	263	**hydroquinone**	785	**ribose**	10975657
butanone	6569	**hydroxychloroquine**	3652	**rutin**	5280805
butyric acid	264	**hypoxanthine**	135398638	**sebacic acid**	5192
caffeine	2519	**ibuprofen**	3672	**serine**	5951
camphor	2537	**imatinib**	5291	**serotonin**	5202
capsaicin	1548943	**imidazole**	795	**sertraline**	68617
carbazole	6854	**indene**	7219	**sildenafil**	135398744
carbon dioxide	280	**indigo**	10215	**simvastatin**	54454
carbon tetrachloride	5943	**indole**	798	**styrene**	7501
carvone	7439	**iodobenzene**	11575	**succinic acid**	1110
catechol	289	**isoleucine**	6306	**succinimide**	11439
chlorobenzene	7964	**isoquinoline**	8405	**sucrose**	5988
chloroethane	6337	**kaempferol**	5280863	**sulfur dioxide**	1119

Molecule	CID	**Molecule**	CID	**Molecule**	CID
chloroform	6212	**lactose**	440995	**tadalafil**	110635
chloroquine	2719	**leucine**	6106	**tamoxifen**	2733526
cholesterol	5997	**lidocaine**	3676	**taurine**	1123
cimetidine	2756	**limonene**	22311	**terephthalic acid**	7489
ciprofloxacin	2764	**linalool**	6549	**testosterone**	6013
citronellol	8842	**lorazepam**	3958	**tetracycline**	54675776
clonazepam	2802	**lovastatin**	53232	**theobromine**	5429
clopidogrel	60606	**lysine**	5962	**theophylline**	2153
cocaine	446220	**maleic anhydride**	7923	**thiamine**	1130
codeine	5284371	**maltose**	439186	**thiazole**	9256
colchicine	6167	**mannose**	18950	**thiophene**	8030
corannulene	11831840	**melatonin**	896	**threonine**	6288
coronene	9115	**menthol**	1254	**thymidine**	5789
cortisol	5754	**metformin**	4091	**thymine**	1135
creatine	586	**methane**	297	**thymol**	6989
creatinine	588	**methanol**	887	**thyroxine**	5819
cubane	136090	**methionine**	6137	**toluene**	1140
curcumin	969516	**methyl acetate**	6584	**tramadol**	33741
cyclohexanol	7966	**methylamine**	6329	**trichlorobenzene**	6895
cyclohexylamine	7965	**metronidazole**	4173	**triethylamine**	8471
cysteine	5862	**morphine**	5288826	**trimethylamine**	1146
cytidine	6175	**morpholine**	8083	**tryptophan**	6305

Molecule	CID	**Molecule**	CID	**Molecule**	CID
cytosine	597	**naphthalene**	931	**tyrosine**	6057
deoxyribose	9828112	**naringenin**	439246	**uracil**	1174
dexamethasone	5743	**niacin**	938	**urea**	1176
diazepam	3016	**nicotine**	89594	**uric acid**	1175
dichloromethane	6344	**norepinephrine**	439260	**uridine**	6029
diethyl ether	3283	**octane**	356	**valine**	6287
diethylamine	8021	**omeprazole**	4594	**venlafaxine**	5656
dimethylamine	674	**oxazole**	9255	**vitamin c**	54670067
dopamine	681	**pantothenic acid**	6613	**warfarin**	54678486
ephedrine	9294	**paracetamol**	1983	**water**	962
epinephrine	5816	**paroxetine**	43815	**xanthine**	1188
estradiol	5757	**penicillin**	2349		
ethane	6324	**pentane**	8003		

Table B.1 – All 256 unique molecules in the sample dataset with their PubChem Compound IDs (CIDs). Molecules were selected to ensure diversity across chemical categories including inorganic compounds, aliphatic organics, aromatics, heteroaromatics, halogenated compounds, amino acids, sugars, nucleobases, vitamins, neurotransmitters, pharmaceutical, and natural products.

Appendix C - SRD Worked Example: Water (H_2O, CID 962)

This appendix presents a manual computation of the Sorted Radial Distance (SRD) invariant for water, verifying the correctness of the computeSRD implementation described in Section 4.2.

C.1

The following coordinates were extracted from the PubChem SDF file for water (CID 962), generated using the MMFF94 force field.

Atoms	X(Å)	y(Å)	z(Å)
O	0.0000	0.0000	0.0000
H_1	0.2774	0.8929	0.2544

H_2	0.6068	-0.2383	-0.7269

Table C.1 – the 3D coordinates for Water on the PubChem dataset.

C.2

The geometric centre is computed as the mean of all atomic coordinates:

$\bar{x} = (0.0000 + 0.2774 + 0.6068) / 3 = 0.2947 \text{ Å}$

$\bar{y} = (0.0000 + 0.8929 + (-0.2383)) / 3 = 0.2182 \text{ Å}$

$\bar{z} = (0.0000 + 0.2544 + (-0.7169)) / 3 = -0.1542 \text{ Å}$

Geometric centre = (0.2947, 0.2182, −0.1542)

C.3

$d(O) = \sqrt{(0.0000 - 0.2947)^2 + (0.0000 - 0.2182)^2 + \big(0.0000 - (-0.1542)\big)^2} =$
$\sqrt{0.0868 + 0.0476 + 0.0238} = \sqrt{0.1582} = \mathbf{0.3977} \text{ Å}$

$d(H^1) = \sqrt{(0.2774 - 0.2947)^2 + (0.8929 - 0.2182)^2 + \big(0.2544 - (-0.1542)\big)^2} =$
$\sqrt{0.0003 + 0.4552 + 0.1670} = \sqrt{0.6225} = \mathbf{0.7890} \text{ Å}$

$d(H^2) = \sqrt{(0.6068 - 0.2947)^2 + (-0.2383 - 0.2182)^2 + \big(-0.7169 - (-0.1542)\big)^2} =$
$\sqrt{0.0974 + 0.2084 + 0.3166} = \sqrt{0.6224} = \mathbf{0.7890} \text{ Å}$

When the above Euclidean distances are sorted in descending order:

$$SRD(H_2O) = [0.7890, 0.7890, 0.3977] \text{ Å}$$

NOTE: The near-symmetric geometry of water is observed in the two hydrogen atoms being equidistant from the geometric centre.

C.4

```
>> conf = mol.GetConformer()
>> atoms = []
>> for atom in mol.GetAtoms():
>>     pos = conf.GetAtomPosition(atom.GetIdx())
>>     atoms.append([pos.x, pos.y, pos.z])
>>
>> coords = np.array(atoms)
>> centre = coords.mean(axis=0)
>> distances = np.sqrt(((coords - centre) ** 2).sum(axis=1))
>> srd = sorted(distances, reverse=True)
>> print([round(x, 4) for x in srd])
>> "
[np.float64(0.789), np.float64(0.789), np.float64(0.3978)]
```

Figure C.1 *– an inline code snippet and its outputs that mimics the ComputeSRD functions.*

The computeSRD function returns $[0.7890, 0.7890, 0.3978]$ Å. The manually computed result matches the code output to four decimal places.

IMAGE ACKNOWLEDGEMENTS

Molecule Line Icon- Vecteezy (https://www.vecteezy.com/vector-art/2238029-molecule-line-icon-on-white)

File Icon- Vecteezy (https://www.vecteezy.com/vector-art/17315027-file-digital-document-line-icon-vector-illustration)

PubChem logo- Wikimedia Customs
https://commons.wikimedia.org/wiki/File:PubChem_logo.svg

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